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Hatano

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(54) **ACTUATOR**

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10,003,746 B2 * 6/2018 Minamisawa G03B 5/00
10,315,221 B2 * 6/2019 Mao B06B 1/045
10,447,130 B2 * 10/2019 Mao B06B 1/045
10,447,131 B2 * 10/2019 Mao H02K 33/16
10,447,134 B2 * 10/2019 Mao H02K 33/02
10,596,594 B2 * 3/2020 Ling H02K 33/02
10,674,278 B2 * 6/2020 Zhou H04R 9/025
10,951,104 B2 * 3/2021 Takeda H02K 41/031
11,201,531 B2 * 12/2021 Jin H02K 33/16

(Continued)

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FOREIGN PATENT DOCUMENTS

JP 6194288 9/2017
JP 2020102901 7/2020
WO 2017002950 1/2017

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(56) **References Cited**

U.S. PATENT DOCUMENTS

9,774,236 B2 * 9/2017 Xu H02K 33/04
9,948,170 B2 * 4/2018 Jun H02K 33/00

OTHER PUBLICATIONS

“Search Report of Europe Counterpart Application”, issued on Oct. 20, 2023, pp. 1–10.

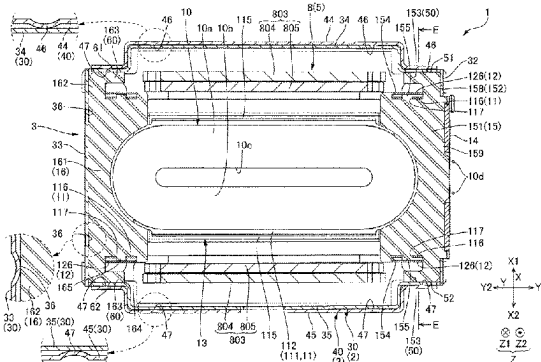
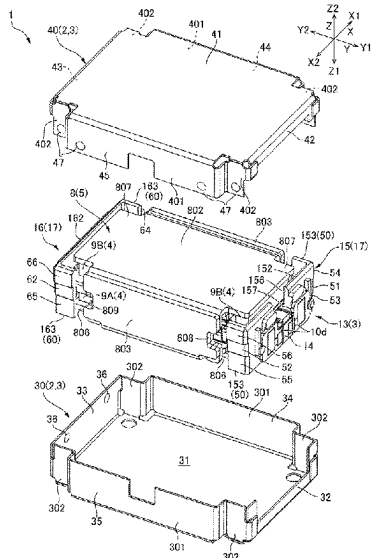
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(57) **ABSTRACT**

This actuator includes a movable body provided with a magnet and a yoke, and a support body provided with a case and a coil that vibrates the movable body. The case includes a first case member having a first end plate portion opposing the movable body from a Z1 direction, and a second case member having a second end plate portion opposing the movable body from a Z2 direction. The first case member includes a first case first side plate portion extending toward the Z2 direction from a Y1 direction end of the first end plate portion, and a first case second side plate portion extending toward the Z2 direction from a Y2 direction end of the first end plate portion. The first case second side plate portion includes a first projection portion. The coil holder is press-fitted between the first projection portion and first case first side plate portion.

9 Claims, 12 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

11,716,003 B1 * 8/2023 Miesner H02K 33/16
 310/27
 11,909,273 B2 * 2/2024 Hatano H01F 7/081
 2009/0267423 A1 * 10/2009 Kajiwara H02K 33/02
 310/38
 2012/0170792 A1 * 7/2012 Li H04R 9/066
 381/412
 2013/0313919 A1 * 11/2013 Nakamura H02K 33/00
 310/25
 2016/0372998 A1 * 12/2016 Xu H02K 33/16
 2017/0033627 A1 * 2/2017 Xu H02K 1/34
 2017/0144191 A1 * 5/2017 Mao B06B 1/045
 2017/0288525 A1 * 10/2017 Lee G03B 3/10
 2017/0310203 A1 * 10/2017 Takeda H02K 33/16
 2017/0373578 A1 * 12/2017 Wu H02K 33/16
 2018/0059794 A1 * 3/2018 Nakamura H02K 33/02
 2018/0248458 A1 * 8/2018 Amemiya H02K 55/04
 2018/0297061 A1 * 10/2018 Mao H02K 5/04
 2018/0297070 A1 * 10/2018 Xu B06B 1/045
 2018/0297071 A1 * 10/2018 Xu B06B 1/045

2018/0297077 A1 * 10/2018 Chai B06B 1/045
 2018/0351442 A1 * 12/2018 Liu H02K 33/16
 2019/0081544 A1 * 3/2019 Zhu H02K 33/02
 2019/0190365 A1 * 6/2019 Takeda H02K 33/18
 2019/0305630 A1 * 10/2019 Mori H01F 27/2823
 2019/0305658 A1 * 10/2019 Mori H02K 33/12
 2019/0305660 A1 * 10/2019 Tsuchihashi H02K 33/16
 2019/0315070 A1 * 10/2019 Yanagisawa H02K 33/16
 2020/0044526 A1 * 2/2020 Tang H02K 33/18
 2020/0044541 A1 * 2/2020 Takeda H02K 33/16
 2020/0059147 A1 * 2/2020 Kitahara H02K 33/16
 2020/0107134 A1 * 4/2020 Mori H04R 7/18
 2020/0161955 A1 * 5/2020 Kitahara H02K 33/16
 2020/0204054 A1 * 6/2020 Mori H02K 33/02
 2021/0075306 A1 * 3/2021 Little H02K 33/02
 2021/0359583 A1 * 11/2021 Toyota H02K 33/16
 2022/0200428 A1 * 6/2022 Shao H02K 33/02
 2022/0200433 A1 * 6/2022 Mao H02K 33/18
 2022/0209634 A1 * 6/2022 Ma H02K 33/02
 2022/0247295 A1 * 8/2022 Takei H02K 9/223
 2022/0352801 A1 * 11/2022 Kitahara H02K 33/02
 2022/0368206 A1 * 11/2022 Ma H02K 33/02
 2023/0387771 A1 * 11/2023 Hatano H02K 1/34

* cited by examiner

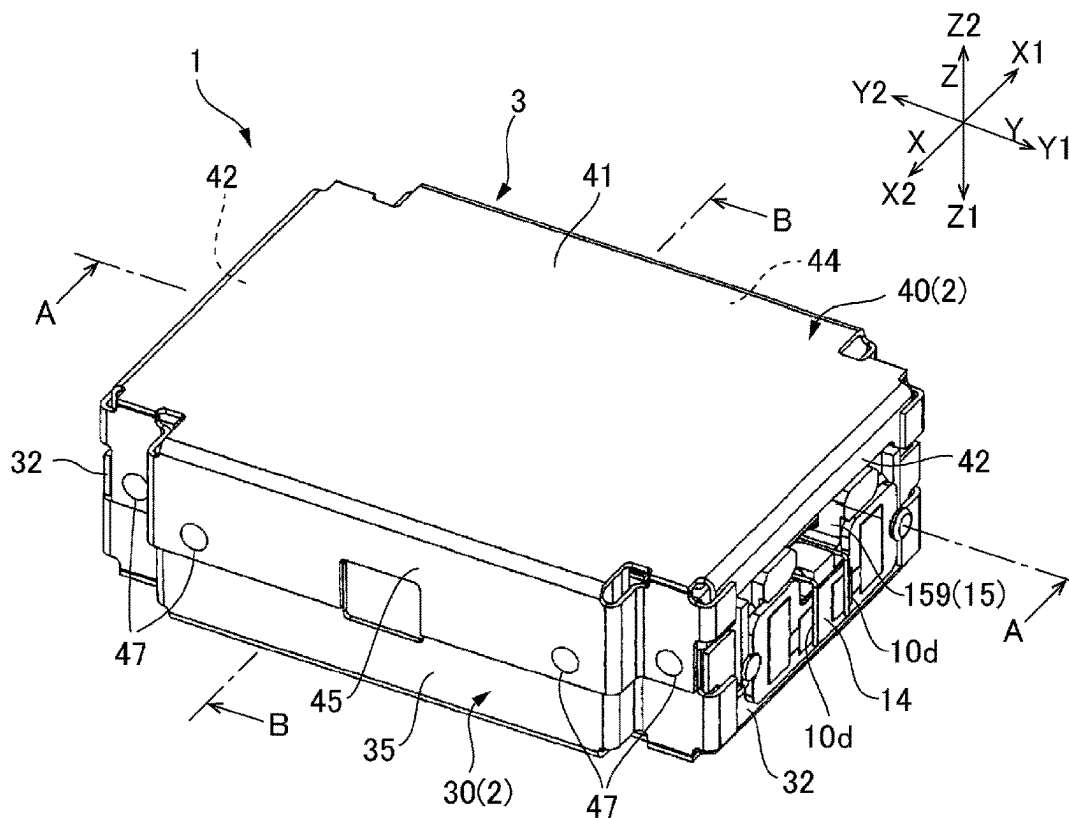


FIG. 1A

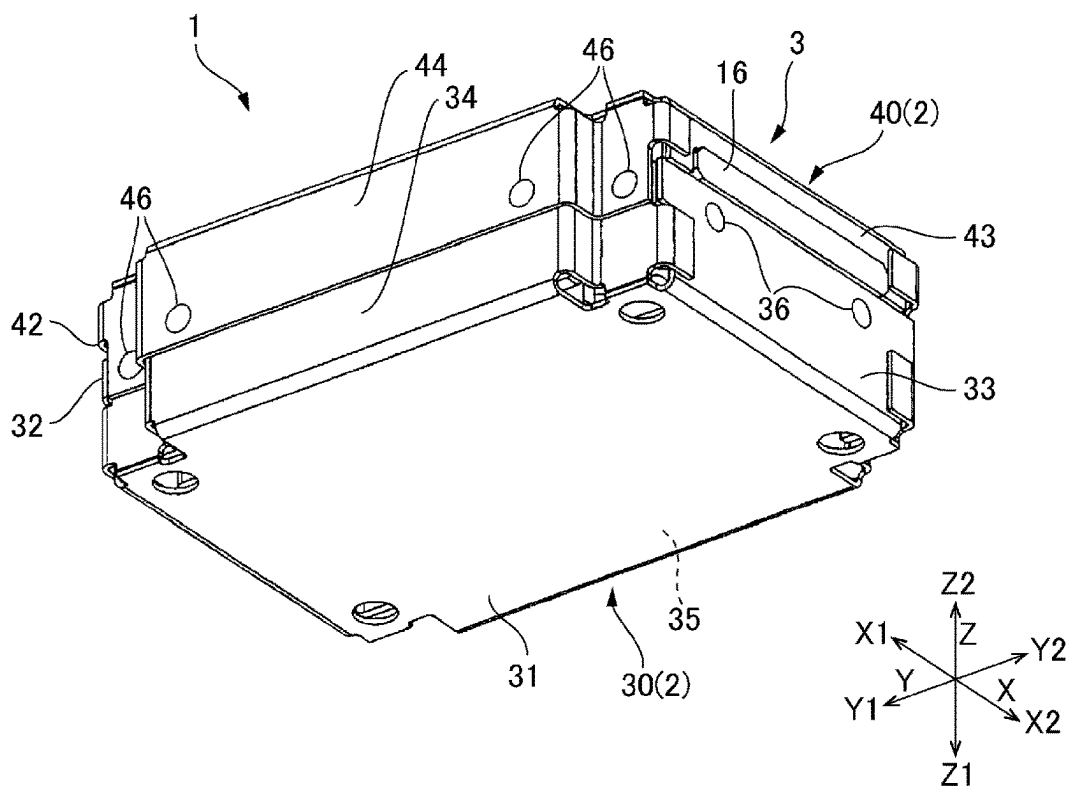


FIG. 1B

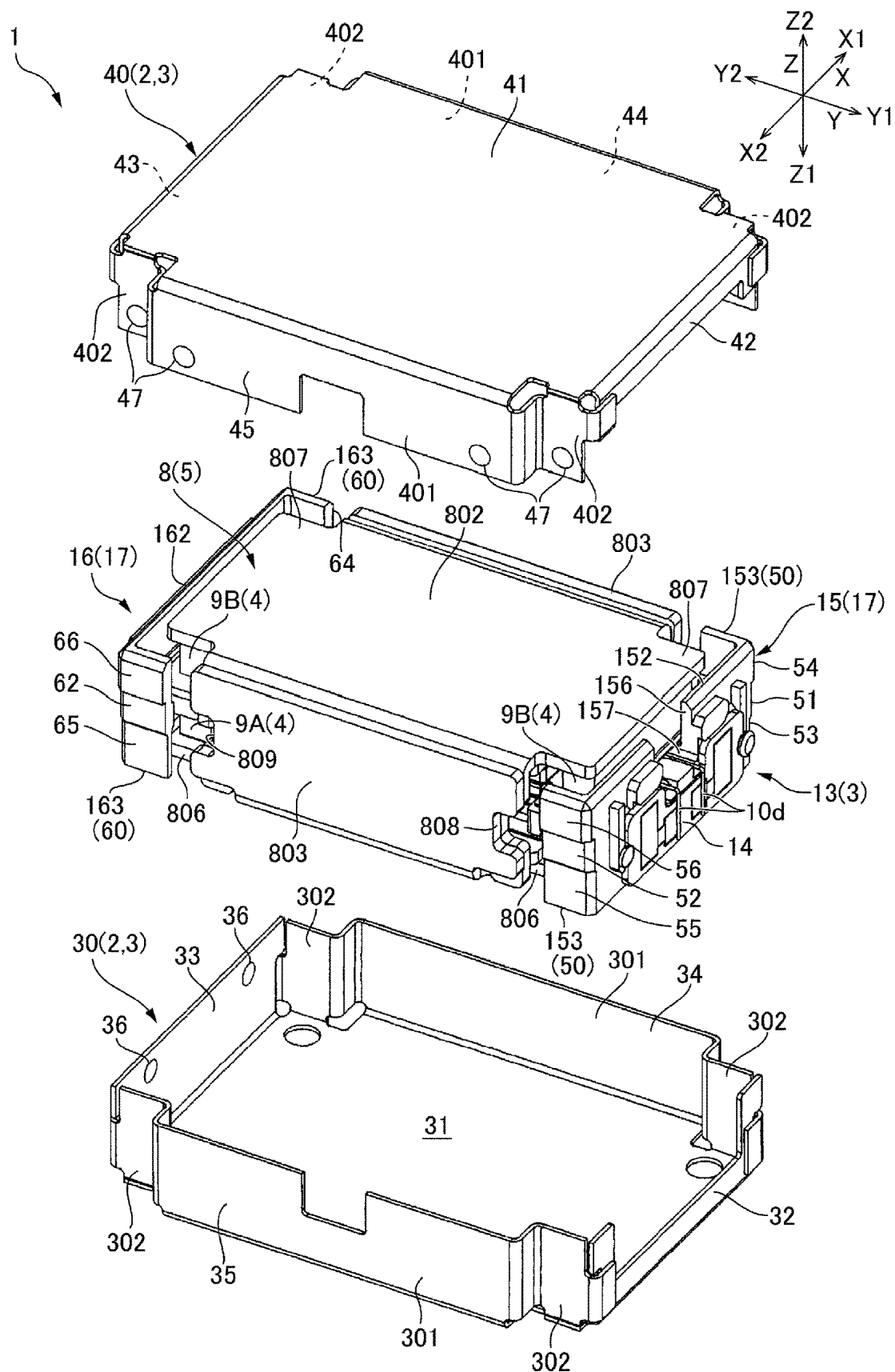


FIG. 2

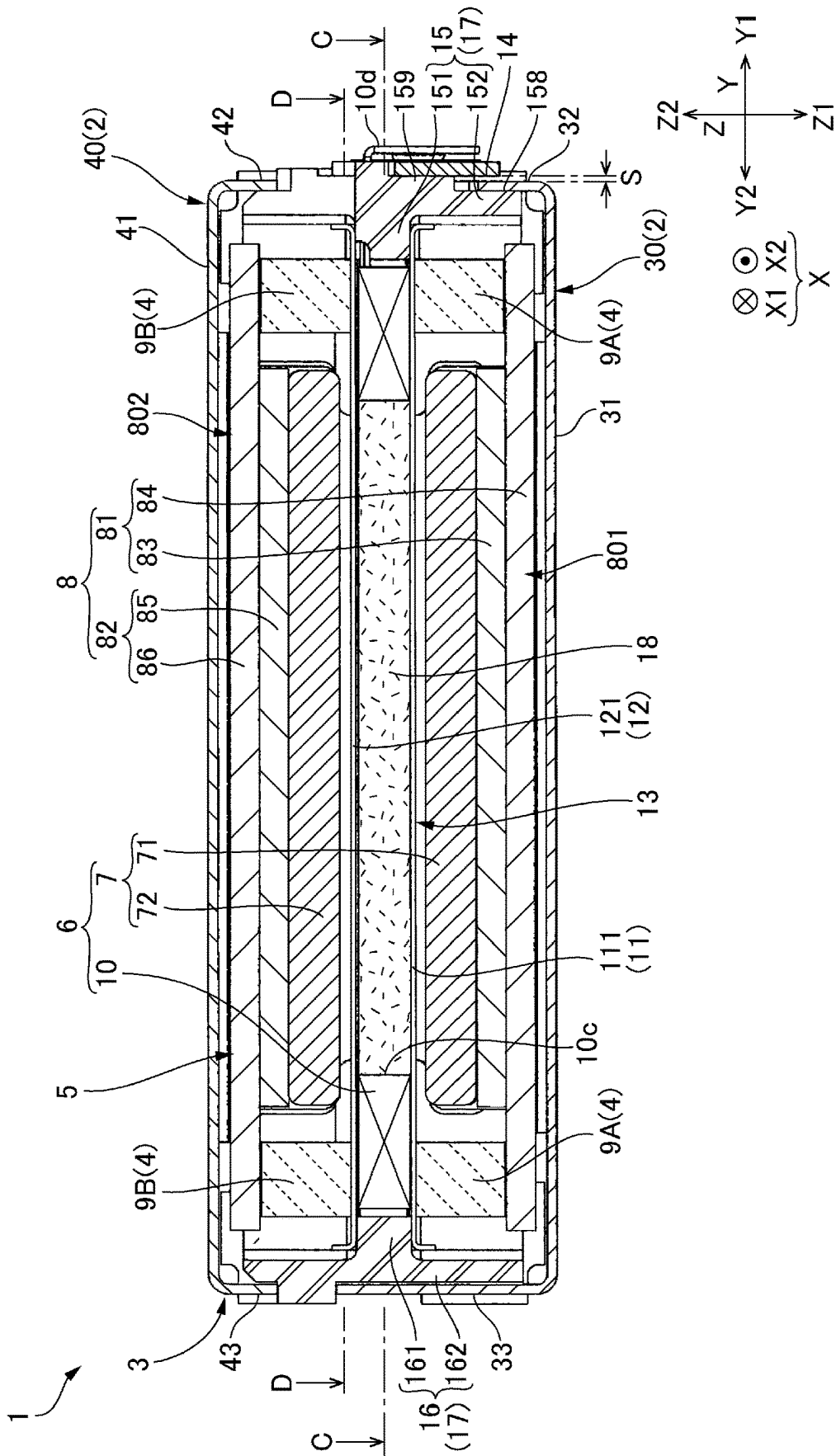


FIG. 3

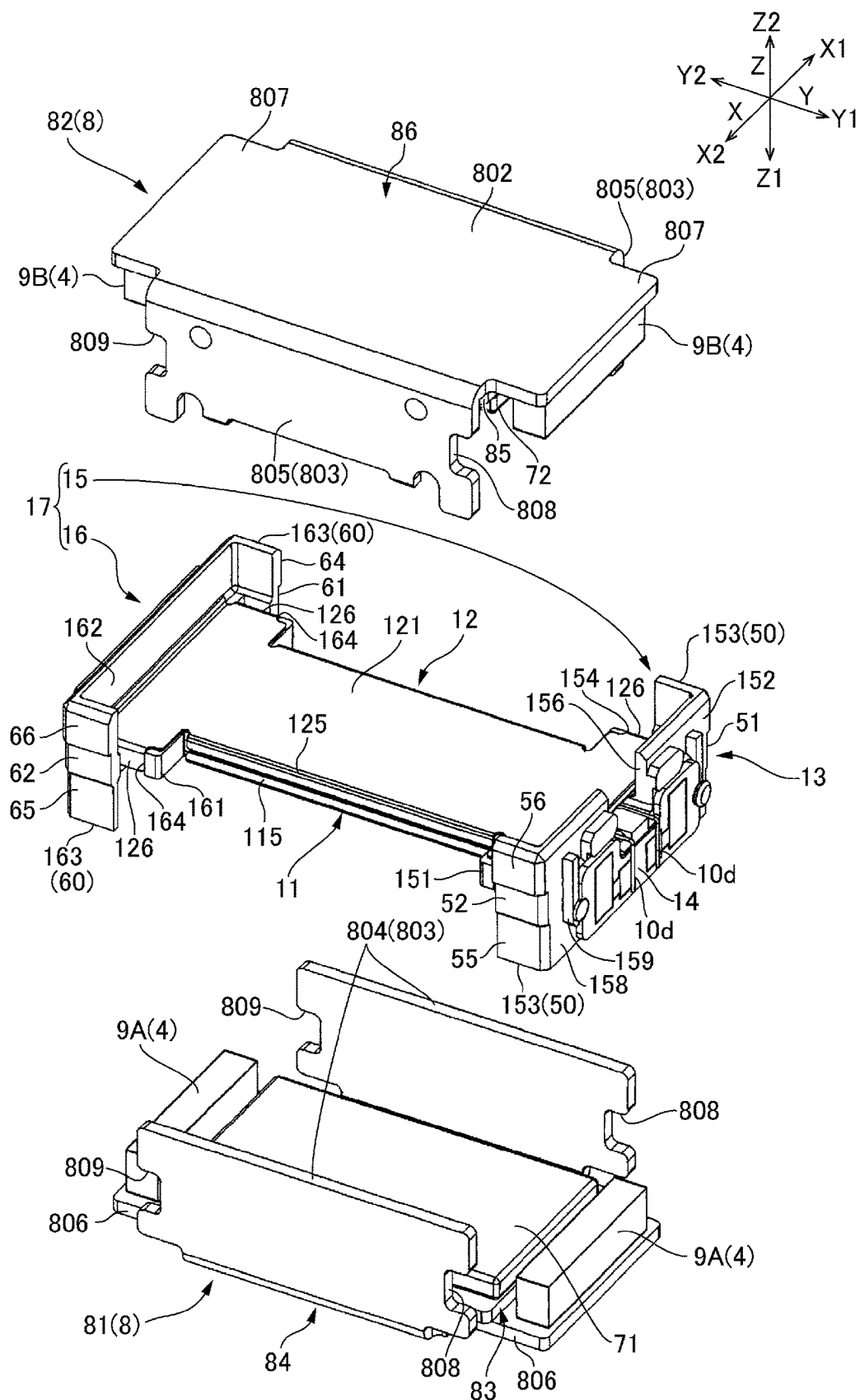


FIG. 5

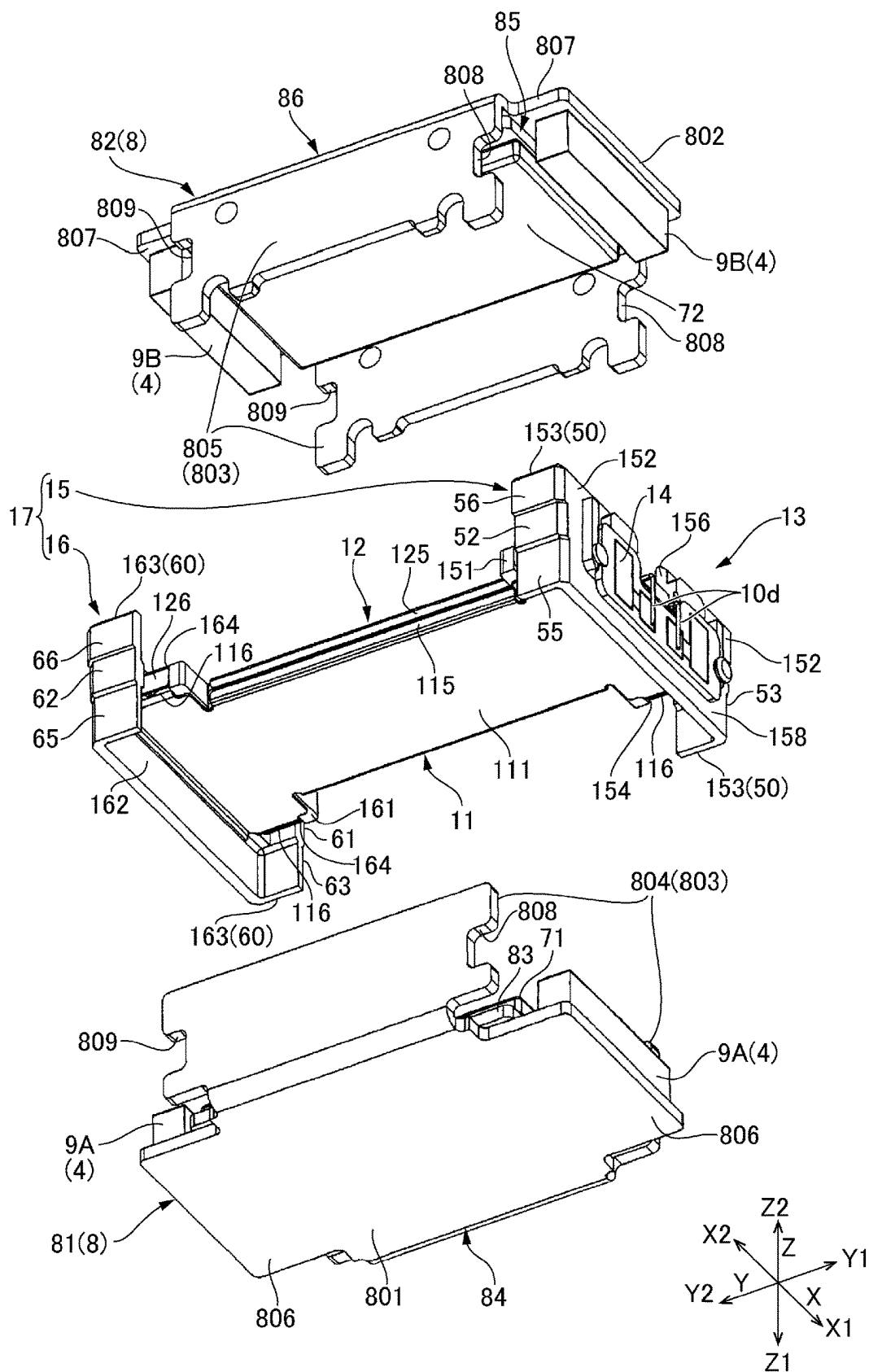


FIG. 6

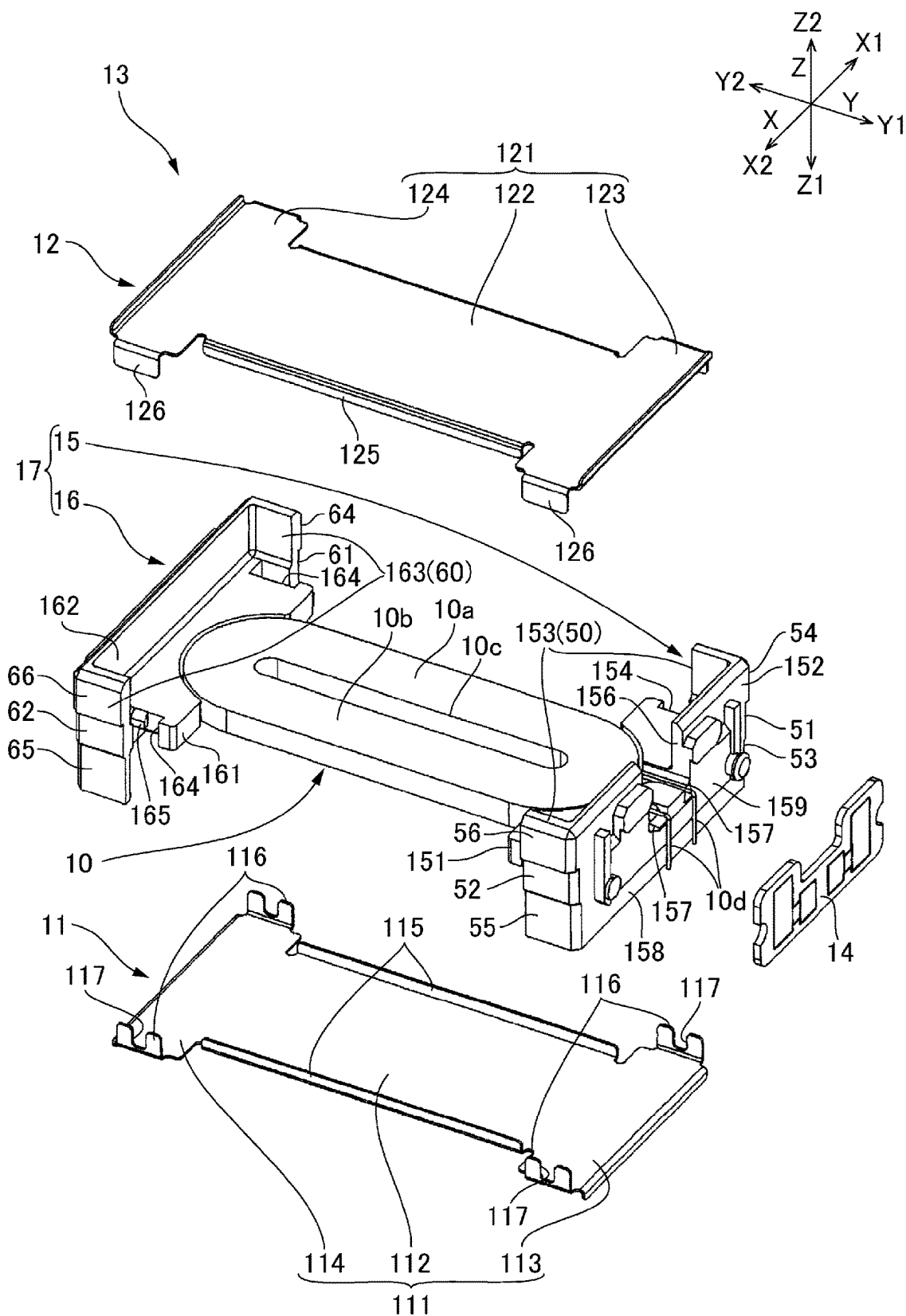


FIG. 7

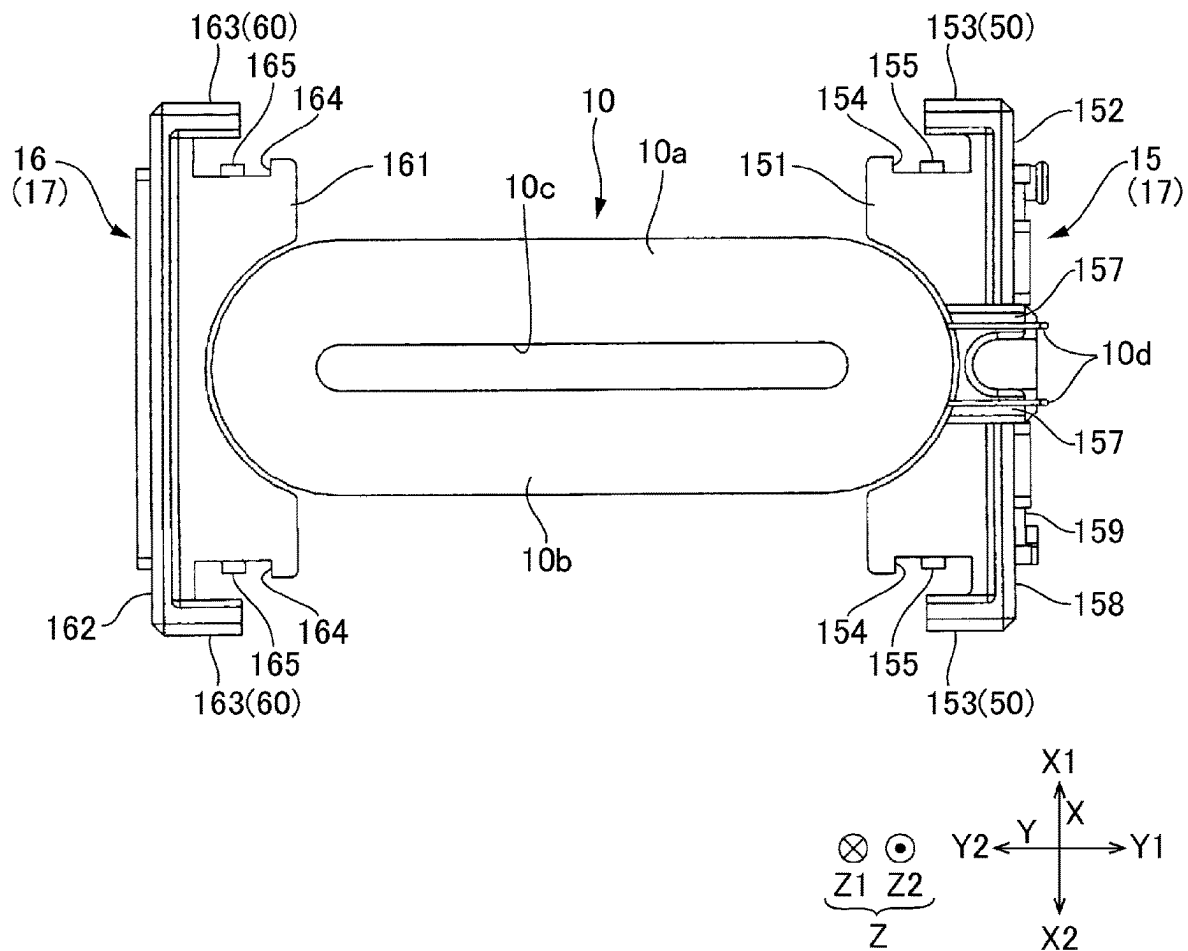
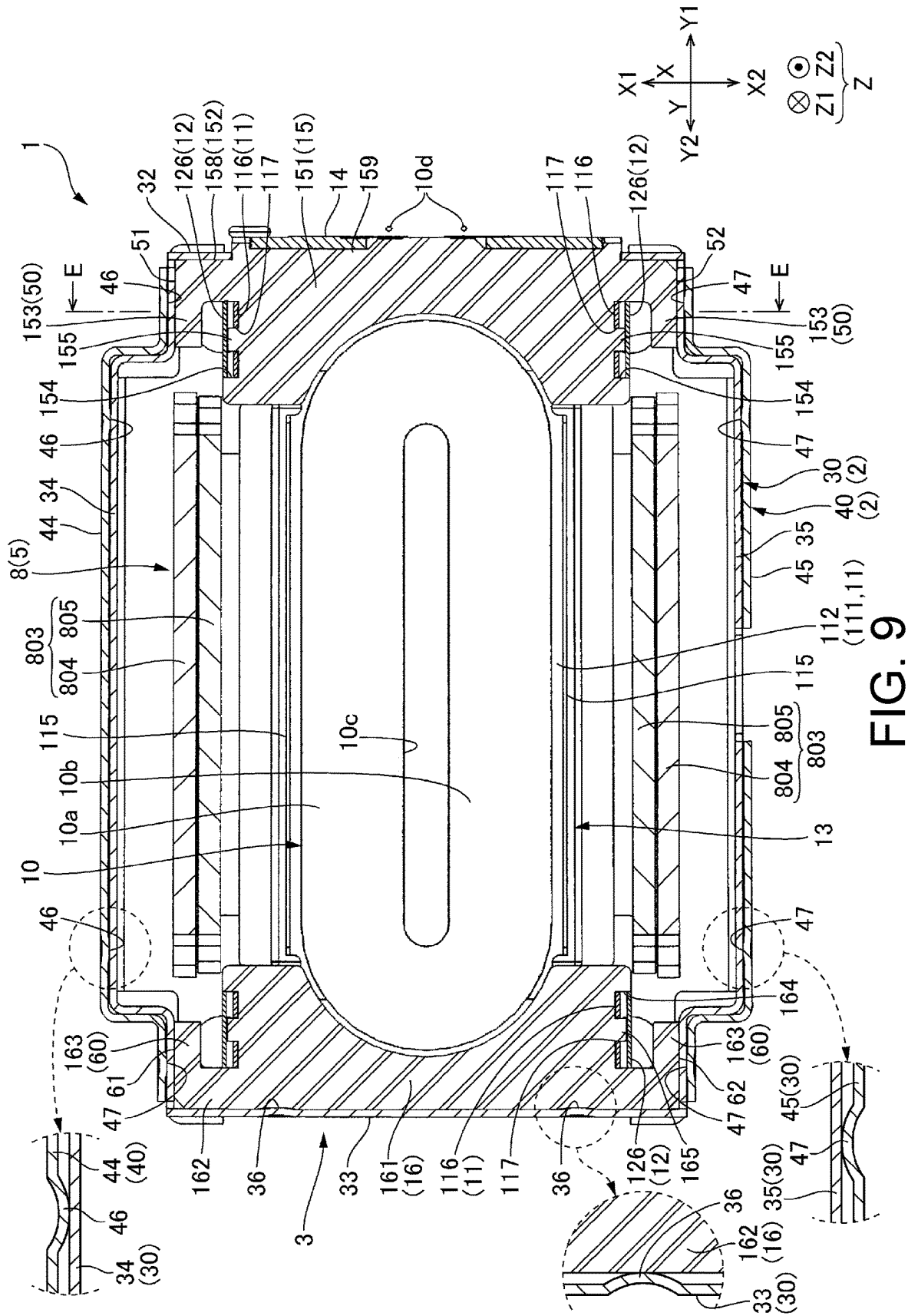


FIG. 8



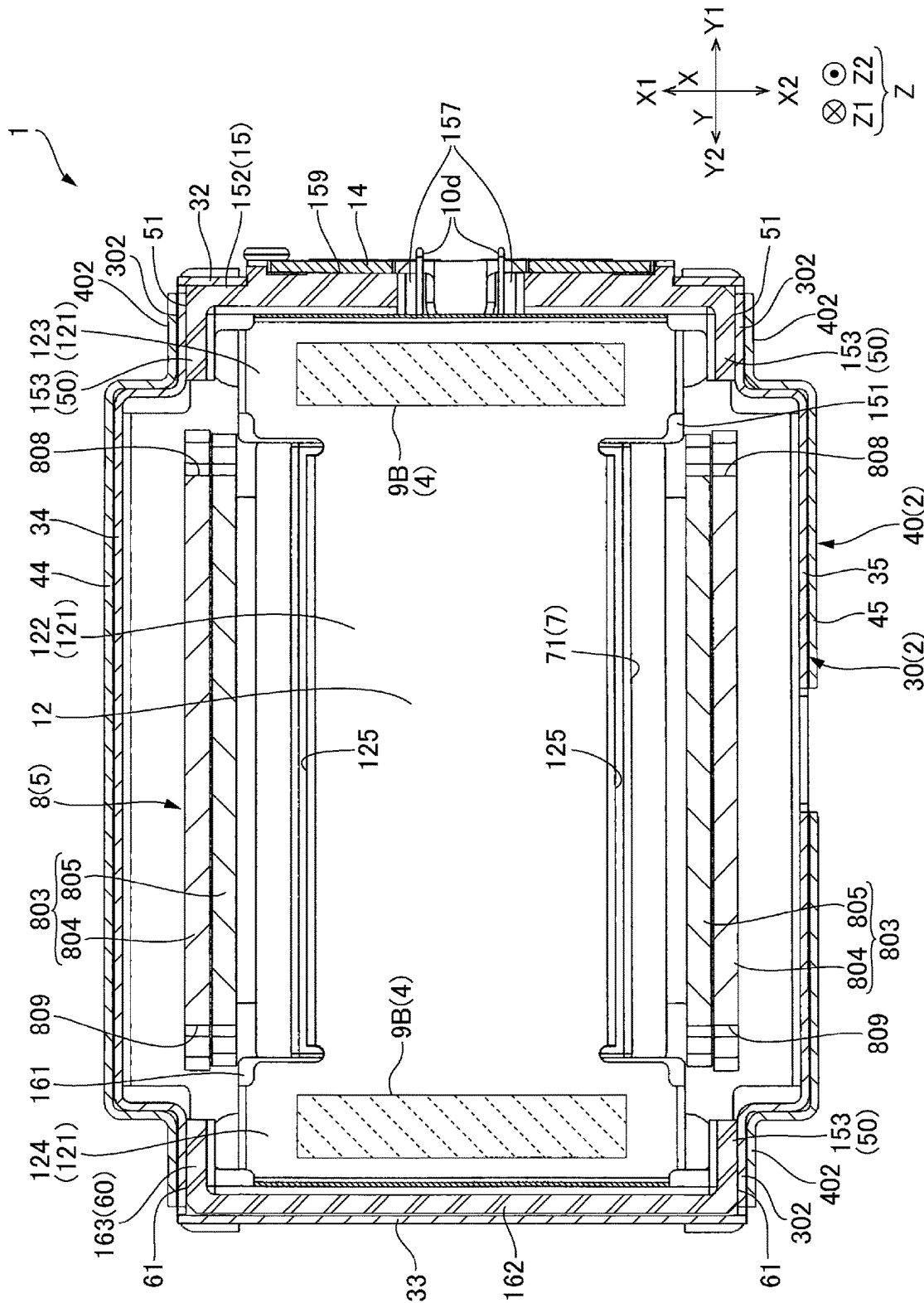


FIG. 11

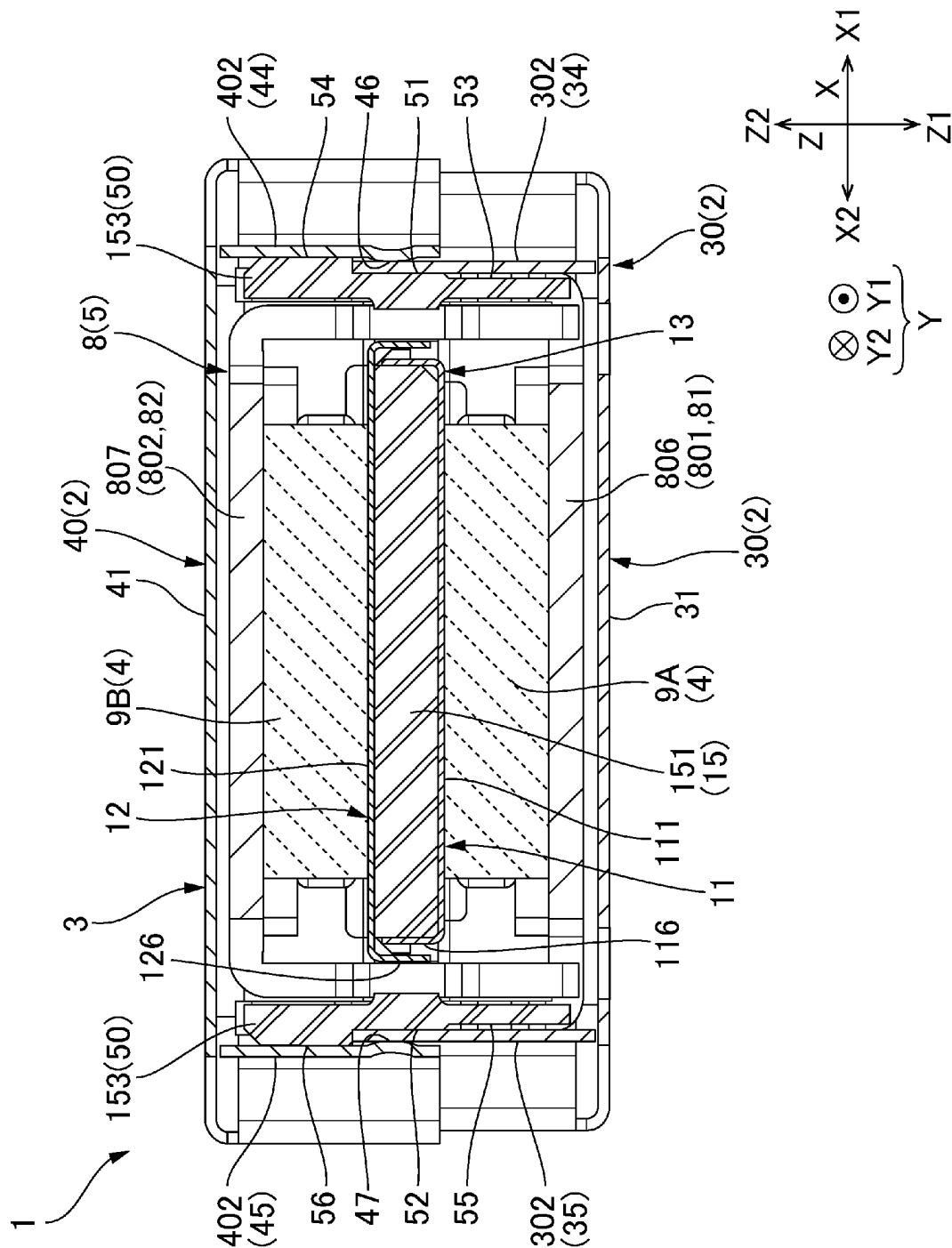


FIG. 12

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ACTUATOR**CROSS-REFERENCE TO RELATED APPLICATION**

This application claims the priority benefit of Japan application serial no. 2022-086491, filed on May 27, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND**Field of the Invention**

At least an embodiment of the present invention relates to an actuator that vibrates a movable body.

Description of the Related Documents

Japanese Patent Application Laid-Open No. 2020-102901 (JPA 2020-102901) discloses an actuator which is provided with a movable body including magnets, and a support body including a coil, and vibrates the movable body with respect to the support body by applying a drive current to the coil. This type of actuator uses an elastic body or a viscoelastic body as a connecting body which connects the support body and the movable body. When the movable body is vibrated, a reaction force corresponding to the vibration of the movable body is applied to the support body through the connecting body. As a result, a user who is touching the support body can feel the vibrations.

In the actuator of JP2020-102901, the support body includes a metal case that defines the outer shape of the actuator, and a coil holder made of resin. The coil is an air core coil and is disposed in a coil placement hole in the coil holder. A metallic plate is attached to the coil holder to cover, from both sides, the coil and a plate portion provided with the coil placement hole. The movable body includes a first yoke that opposes the coil from one side, and a second yoke that opposes the coil from an other side, and magnets are fixed to each of the first yoke and the second yoke. The connecting body is disposed at both ends of the movable body in a longitudinal direction in locations where the yoke and the plate covering the coil oppose each other.

The case of JP2020-102901 includes a first case member that covers the coil holder from one side, and a second case member that covers the coil holder from an other side. The coil holder includes side plate portions formed on an outer edge of the plate portion provided with the coil placement hole. The case and the coil holder are assembled by fitting the side plate portions of the coil holder inside the first case member and the second case member, which are manufactured by bending metal plates.

In JP2020-102901, the first case member and the second case member are provided with engagement holes. The coil holder and the case are assembled by fitting engagement protrusion portions provided on the side surfaces of the coil holder into the engagement holes in each case member.

Here, when assembling the coil holder inside the metal case, the dimensions are set such that a gap is formed between the case and the coil holder in order to ensure assembly even when there is a tolerance in the dimensions of the components. Therefore, the coil holder may move inside the case during vibration, which generates noise. In order to prevent this, a structure has been proposed in which an adhesive is inserted into the gap between the case and the

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coil holder such that they are fixed. However, the number of assembly steps may increase due to an additional injection step and curing step of the adhesive.

Accordingly, at least an embodiment of the present invention is to suppress rattling of the coil holder against the case during vibration to reduce noise, and to suppress an increase in the number of assembly steps.

SUMMARY

An actuator according to at least an embodiment of the present invention includes: a movable body; a support body provided with a metallic case that accommodates the movable body, and a coil holder made of resin; a connecting body that is connected to the movable body and the support body; and a magnetic drive circuit which is provided with a coil that is held by the coil holder, and a magnet that opposes the coil in a first direction, and which vibrates the movable body with respect to the support body in a second direction that intersects the first direction; wherein the case is provided with a first case member having a first end plate portion that opposes the movable body from one side in the first direction, and a second case member having a second end plate portion that opposes the movable body from an other side in the first direction, the first case member is provided with a first case first side plate portion that extends toward the other side in the first direction from an end on one side of the first end plate portion in a third direction, and a first case second side plate portion that extends toward the other side in the first direction from an end on an other side of the first end plate portion in the third direction, the third direction being a direction that intersects the first direction and intersects the second direction, the first case second side plate portion is provided with a first projection portion that protrudes toward the one side in the third direction, and the coil holder is press-fitted between the first projection portion and the first case first side plate portion.

According to at least an embodiment of the present invention, the first projection portion is provided on the side plate portion (first case second side plate portion) on the other side of the first case member in the third direction. Therefore, when the case and the coil holder are assembled, the first projection portion can be brought into elastic contact with the coil holder. Further, the coil holder can be positioned as a result of being biased toward the side plate portion (first case first side plate portion) on the one side in the third direction. Consequently, rattling of the coil holder inside the case can be suppressed, and the noise due to rattling of the coil holder can be suppressed. Furthermore, even when the dimensional accuracy of the first case member is low, the coil holder can be brought into contact with the first case first side plate portion with certainty, which provides good ease of assembly. In addition, because it is not necessary to fix the coil holder to the case using an adhesive, an injection step and curing step of the adhesive is not necessary. Therefore, an increase in the number of assembly steps can be suppressed.

In at least an embodiment of the present invention, in the first projection portion, a surface on the other side in the third direction has a concave shape, and a surface on the one side in the third direction is a half punched portion having a convex shape. In this way, the first case member provided with the first projection portion can be easily manufactured. Therefore, an increase in component costs can be suppressed.

In at least an embodiment of the present invention, the coil holder comprises a first holder member provided with a

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first coil holding portion disposed on the one side of the coil in the third direction, and a first holder side plate portion that extends in the first direction from an end of the first coil holding portion on the one side in the third direction, and a second holder member provided with a second coil holding portion disposed on the other side of the coil in the third direction, and a second holder side plate portion that extends in the first direction from an end of the second coil holding portion on the other side in the third direction, the first holder member and the second holder member are fixed via a metallic first plate, which overlaps the first coil holding portion and the second coil holding portion from the one side in the first direction, and a metallic second plate, which overlaps the first coil holding portion and the second coil holding portion from the other side in the first direction, the first projection portion makes contact with the second holder side plate portion from the other side in the third direction, and the first holder side plate portion makes contact with the first case first side plate portion from the other side in the third direction. As a result, by dividing the coil holder into two members by dividing it in the Y direction, a resin part that covers both sides of the coil in the X direction becomes unnecessary. Therefore, the external dimensions of the actuator in the X direction can be reduced. Alternatively, the coil can be made larger without increasing the size of the outer shape of the actuator, and the thrust of the magnetic drive circuit can be increased to generate large vibrations. Furthermore, because the coil can be protected by the first plate and the second plate, the possibility that the coil will be damaged due to collision with the magnet or collision with the yoke is low. In addition, the first holder member and the second holder member can be handled as the coil assembly as a result of being fixed via the first plate and the second plate. Therefore, even when the coil holder is divided into two members, a reduction in the ease of assembly of the actuator can be suppressed, and it is also possible for the first projection portion to be brought into elastic contact and positioned with respect to the coil holder.

In at least an embodiment of the present invention, the coil holder is provided with a contact surface that makes contact with the first case first side plate portion from the other side in the third direction, and a substrate fixing portion that protrudes from the contact surface toward the one side of the first case first side plate portion in the third direction. In this way, a gap is formed in the third direction between the power feed substrate that is fixed to the substrate fixing portion, and the first case first side plate portion. Therefore, because contact between the metallic case and the power feed substrate can be avoided, it is possible to prevent the pattern on the power feed substrate and the case from making contact and short-circuiting.

In at least an embodiment of the present invention, the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction, the second case member is provided with a second case third side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the one side in the second direction, and a second case fourth side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the other side in the second direction, and the first case third side plate portion and the first case fourth side plate portion are press-fitted

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between the second case third side plate portion and the second case fourth side plate portion. In this way, because the first case member and the second case member can be positioned by press-fitting, they do not need to be fixed using an adhesive. Therefore, an increase in the number of assembly steps can be suppressed.

In at least an embodiment of the present invention, one of the second case third side plate portion and the first case third side plate portion is provided with a second projection portion that protrudes toward the other of the second case third side plate portion and the first case third side plate portion, and one of the second case fourth side plate portion and the first case fourth side plate portion is provided with a third projection portion that protrudes toward the other of the second case fourth side plate portion and the first case fourth side plate portion. In this way, a press-fitted state can be formed by bringing the leading edge of the second projection portion into elastic contact with the first case third side plate portion, and the leading edge of the third projection portion into elastic contact with the first case fourth side plate portion. Therefore, press-fitting can be performed with certainty even when the dimensional accuracy of the first case member and the second case member is low, and a reduction in the ease of assembly can also be suppressed.

In at least an embodiment of the present invention, the second projection portion and the third projection portion are half punched portions. In this way, the second case member **40** provided with the second projection portion **46** and the third projection portion **47** can be easily manufactured because they can be easily formed by a method such as press working. Therefore, an increase in component costs can be suppressed.

In at least an embodiment of the present invention, the movable body is provided with a yoke that holds the magnet, the yoke is provided with a first opposing portion that opposes the coil from the one side in the first direction, a second opposing portion that opposes the coil from the other side in the first direction, and a pair of connection portions that extend in the first direction on both sides of the coil in the second direction and connect the first opposing portion and the second opposing portion, the coil holder is provided with a pair of first holder projection portions that protrude toward the other side in the third direction from edges on both sides of the first holder side plate portion in the second direction, and a pair of second holder projection portions that protrude toward the one side in the third direction from edges on both sides of the second holder side plate portion in the second direction, an end portion of the first opposing portion and the second opposing portion on the one side in the third direction is disposed between the pair of first holder projection portions, and an end portion of the first opposing portion and the second opposing portion on the other side in the third direction is disposed between the pair of second holder projection portions. In this way, the first holder projection portions and the second holder projection portions function as stoppers that restrict the movable range of the movable body in the second direction. Therefore, because stoppers can be provided on the coil holder instead of the case, it is less likely that the movable body will collide with the case and be deformed or destroyed due to an impact caused by falling or the like, and the impact resistance of the actuator can be increased.

In at least an embodiment of the present invention, the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate

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portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction, and the pair of first holder projection portions and the pair of second holder projection portions are press-fitted between the first case third side plate portion and the first case fourth side plate portion. In this way, rattling of the coil holder 17 in the X direction can be suppressed. Furthermore, the first case third side plate portion and the first case fourth side plate portion are in a state where they are always pressed toward the outside by the coil holder. As a result, because the first case third side plate portion and the first case fourth side plate portion are always pressed against the second case third side plate portion and the second case fourth side plate portion from the inside, the first case member 30 can be positioned with respect to the second case member 40 in a state where it does not rattle. Moreover, because the parts that function as stoppers that regulate the movable range of the movable body 5 in the X direction can be utilized for press-fitted fixing, the press-fitted fixing can be firmly performed and the configuration of the coil holder 17 can be simplified.

According to at least an embodiment of the present invention, the first projection portion is provided on the side plate portion (first case second side plate portion) on the other side of the first case member in the third direction. Therefore, when the case and the coil holder are assembled, the first projection portion can be brought into elastic contact with the coil holder. Further, the coil holder can be positioned as a result of being biased toward the side plate portion (first case first side plate portion) on the one side in the third direction. Consequently, rattling of the coil holder inside the case can be suppressed, and the noise due to rattling of the coil holder can be suppressed. Furthermore, even when the dimensional accuracy of the first case member is low, the coil holder can be brought into contact with the first case first side plate portion with certainty, which provides good ease of assembly. In addition, because it is not necessary to fix the coil holder to the case using an adhesive, an injection step and curing step of the adhesive is not necessary. Therefore, an increase in the number of assembly steps can be suppressed.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will now be described, by way of example only, with reference to the accompanying drawings which are meant to be exemplary, not limiting, and wherein like elements are numbered alike in several figures, in which:

FIGS. 1A and 1B are perspective views of an actuator to which at least an embodiment of the present invention is applied when viewed from a Z2 direction and a Z1 direction.

FIG. 2 is an exploded perspective view of an actuator showing a support body disassembled into a coil assembly, a first case member, and a second case member.

FIG. 3 is a cross-sectional view of an actuator taken in the YZ plane (a cross-sectional view taken along position A-A in FIG. 1A).

FIG. 4 is a cross-sectional view of an actuator taken in the XZ plane (a cross-sectional view taken along position B-B in FIG. 1A).

FIG. 5 is an exploded perspective view of a movable body and a coil assembly when viewed from a Z2 direction.

FIG. 6 is an exploded perspective view of a movable body and a coil assembly when viewed from a Z1 direction.

FIG. 7 is an exploded perspective view of a coil assembly.

FIG. 8 is a plan view of a coil holder and a coil.

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FIG. 9 is a cross-sectional view of an actuator taken in the XY plane (a cross-sectional view taken along position C-C in FIG. 3) and partially enlarged views thereof.

FIG. 10 is a cross-sectional view of an actuator showing the plan view shape and arrangement of a connecting body (a cross-sectional view taken along position D-D in FIG. 3).

FIG. 11 is a side view of a movable body, a connecting body, and a coil assembly when viewed from an X2 direction.

FIG. 12 is a cross-sectional view of an actuator taken at the position of a press-fitted fixed portion (a cross-sectional view taken along position E-E in FIG. 9).

DETAILED DESCRIPTION

An embodiment of an actuator to which at least an embodiment of the present invention is applied will now be described with reference to the drawings.

Overall Configuration

FIG. 1A is a perspective view of an actuator 1 to which at least an embodiment of the present invention is applied when viewed from a Z2 direction.

FIG. 1B is a perspective view of the actuator 1 to which at least an embodiment of the present invention is applied when viewed from a Z1 direction. FIG. 2 is an exploded perspective view of the actuator 1, showing a support body 3 disassembled into a coil assembly 13, a first case member 30, and a second case member 40. FIG. 3 is a cross-sectional view of the actuator 1 taken in the YZ plane, and is a cross-sectional view taken along position A-A in FIG. 1A. FIG. 4 is a cross-sectional view of the actuator 1 taken in the XZ plane, and is a cross-sectional view taken along position B-B in FIG. 1A. FIG. 5 is an exploded perspective view of a movable body and the coil assembly 13 when viewed from a Z2 direction. FIG. 6 is an exploded perspective view of a movable body 5 and the coil assembly 13 when viewed from a Z1 direction. FIG. 7 is an exploded perspective view of the coil assembly 13. FIG. 8 is a plan view of a coil holder 17 and a coil 10. FIG. 9 is a cross-sectional view and partially enlarged views of the actuator 1 taken in the XY plane, and is a cross-sectional view taken along position C-C in FIG. 3.

The actuator 1 is used as a tactile device that transmits information by vibrations. As illustrated in FIGS. 1A and 1B, the outer shape of the actuator 1 is substantially in the form of a rectangular parallelepiped. The actuator 1 generates vibrations in a transverse direction of the outer shape. In the description below, the transverse direction in which vibrations are generated will be referred to as the X direction (second direction). The longitudinal direction of the actuator 1, which is orthogonal to the X direction, will be referred to as the Y direction (third direction). The thickness direction (height direction) of the actuator 1, which is orthogonal to the X direction and the Y direction, will be referred to as the Z direction (first direction). Furthermore, the one side in the X direction will be referred to as the X1 direction, and the other side will be referred to as the X2 direction. The one side in the Y direction will be referred to as the Y1 direction, and the other side will be referred to as the Y2 direction. The one side in the Z direction will be referred to as the Z1 direction, and the other side will be referred to as the Z2 direction.

As shown in FIGS. 1A to 4, the actuator 1 includes a support body 3 provided with a case 2 that defines the outer shape of the actuator 1, and a movable body 5 which is accommodated inside the case 2. Furthermore, the actuator 1 is provided with a connecting body 4 which connects the support body 3 and the movable body 5, and a magnetic

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drive circuit 6 which relatively moves the movable body 5 with respect to the support body 3 in the X direction (see FIGS. 3 and 4). As shown in FIG. 3, the connecting body 4 includes first connecting bodies 9A and second connecting bodies 9B.

As shown in FIGS. 1A, 1B, and 2, the support body 3 is provided with a case 2 and a coil assembly 13. As shown in FIGS. 3 and 4, the coil assembly 13 includes a coil 10 disposed in the center of the actuator 1 in the Z direction. The case 2 is provided with a first case member 30 and a second case member 40 that are stacked in the Z direction. The first case member 30 is mounted on the coil assembly 13 from the Z1 direction. The second case member 40 is mounted on the coil assembly 13 and the first case member 30 from the Z2 direction.

As shown in FIGS. 3 and 4, the movable body 5 includes a magnet 7 and a yoke 8. As shown in FIG. 4, the magnet 7 is fixed to an inner surface of the yoke 8, which is configured so as to surround the coil 10. The coil 10 and the magnet 7, which are opposed in the Z direction, constitute the magnetic drive circuit 6. The movable body 5 includes, as the magnet 7, a first magnet 71 and a second magnet 72. The first magnet 71 and the second magnet 72 are polarized into two poles in the X direction. When the movable body 5 and the support body 3 are assembled, the coil 10 opposes the first magnet 71 in the Z1 direction, and opposes the second magnet 72 in the Z2 direction.

Coil Assembly

As shown in FIG. 7, the coil assembly 13 is provided with the coil 10, a first plate 11 stacked in the Z1 direction of the coil 10, and a second plate 12 stacked in the Z2 direction of the coil 10. The first plate 11 and the second plate 12 are made from a non-magnetic metal. Furthermore, the coil assembly 13 is provided with a first holder member 15 disposed on the Y1 side of the coil and a second holder member 16 disposed on the Y2 side of the coil 10. The first holder member 15 and the second holder member 16 constitute a coil holder 17. The first holder member 15 and the second holder member 16 are made from resin.

As shown in FIGS. 2 and 3, the coil 10 is positioned at the center of the case 2 in the Z direction. The coil 10 is a flat air core coil, and the thickness direction thereof is oriented in the Z direction. As shown in FIG. 7, the coil 10 has an oval shape that is long in the Y direction, and includes a pair of long side portions 10a and 10b that extend parallel to each other in the Y direction. A central hole 10c extending in the Y direction is provided between the pair of long side portions 10a and 10b. The long side portions 10a and 10b of the coil 10 oppose the first magnet 71 and the second magnet 72 in the Z direction.

As shown in FIGS. 7 and 8, the first holder member 15 includes a first coil holding portion 151 disposed between the first plate 11 and the second plate 12, a first holder side plate portion 152 that extends from the end of the first coil holding portion 151 on the Y1 side toward the Z1 direction and the Z2 direction, and a pair of first holder projection portions 153 that protrude in the Y2 direction from both ends of the first holder side plate portion 152 in the X direction. The first coil holding portion 151 is provided with recess portions 154 formed on both side surfaces in the X direction, and claw portions 155 that protrude from the center of the recess portions 154 in the Y direction. The pair of first holder projection portions 153 each extend to a position opposing the claw portions 155 in the X direction.

A power feed substrate 14 is fixed to the first holder member 15. In the present embodiment, the power feed substrate 14 is a rigid substrate. However, the power feed

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substrate 14 may be a flexible printed circuit board. Power is supplied to the coil 10 via the power feed substrate 14. Two coil wires 10d that have been drawn out from the coil 10 are disposed in groove portions 157 formed in the first coil holding portion 151, drawn out to the Y1 side of the first holder side plate portion 152 from a notch portion 156, in which a central part of the first holder side plate portion 152 in the X direction is notched toward the Z1 direction, and then bent toward the Z1 side and connected to the power feed substrate 14.

As shown in FIG. 7, the power feed substrate 14 is fixed to a substrate fixing portion 159 that protrudes from a contact surface 158, which is a side surface of the first holder side plate portion 152 in the Y1 direction. The substrate fixing portion 159 surrounds the notch portion 156, through which the coil wires 10d are drawn out, on both sides in the X direction and on the Z1 side. As shown in FIG. 3, the end portion of the power feed substrate 14 on the Z1 side extends to a position further on the Z1 side than the end portion of the substrate fixing portion 159 on the Z1 side.

As shown in FIGS. 7 and 8, the second holder member 16 includes a second coil holding portion 161 disposed between the first plate 11 and the second plate 12, a second holder side plate portion 162 that extends from the end of the second coil holding portion 161 on the Y1 side toward the Z1 direction and the Z2 direction, and a pair of second holder projection portions 163 that protrude in the Y1 direction from both ends of the second holder side plate portion 162 in the X direction. The second coil holding portion 161 is provided with recess portions 164 formed on both side surfaces in the X direction, and claw portions 165 that protrude from the center of the recess portions 164 in the Y direction. The pair of second holder projection portions 163 each extend to a position opposing the claw portions 165 in the X direction.

The coil 10 is disposed between the first coil holding portion 151 and the second coil holding portion 161. As shown in FIG. 7, the first plate 11 includes a plate-shaped first plate portion 111 that overlaps the coil 10 from the Z1 side. The first plate portion 111 includes a central plate portion 112 that overlaps the long side portions 10a and 10b of the coil 10, a one side plate portion 113 that extends in the Y1 direction from the central plate portion 112 and overlaps the first coil holding portion 151, and an other side plate portion 114 that extends in the Y2 direction from the central plate portion 112 and overlaps the second coil holding portion 161. The width of the one side plate portion 113 and the other side plate portion 114 in the X direction is larger than the width of the central plate portion 112 in the X direction.

A pair of central bent portions 115 that are bent in the Z2 direction are provided on both ends of the central plate portion 112 in the X direction. A pair of fixing bent portions 116 that are bent in the Z2 direction are provided on both ends of the one side plate portion 113 in the X direction and both ends of the other side plate portion 114 in the X direction. The fixing bent portions 116 in the four locations are each provided with a rectangular notch portion 117 that is notched toward the Z1 direction.

The second plate 12 includes a flat second plate portion 121 that overlaps the coil 10 from the Z2 side. The second plate portion 121 includes a central plate portion 122 that overlaps the long side portions 10a and 10b of the coil 10, a one side plate portion 123 that extends in the Y1 direction from the central plate portion 122 and overlaps the first coil holding portion 151, and an other side plate portion 124 that extends in the Y2 direction from the central plate portion

122 and overlaps the second coil holding portion 161. The width of the one side plate portion 123 and the other side plate portion 124 in the X direction is larger than the width of the central plate portion 122 in the X direction.

A pair of central bent portions 125 that are bent in the Z1 direction are provided on both ends of the central plate portion 122 in the X direction. As shown in FIGS. 5 and 6, the leading edges of the central bent portions 125 oppose the leading edges of the central bent portions 115 of the first plate 11 in the Z direction. The central bent portions 115 of the first plate 11 and the central bent portions 125 of the second plate 12 cover the side surfaces of the long side portions 10a and 10b of the coil 10 in the X direction. A pair of fixing bent portions 126 that are bent in the Z1 direction are provided on both ends of the one side plate portion 123 in the X direction and both ends of the other side plate portion 124 in the X direction of the second plate 12.

When the coil assembly 13 is assembled, the fixing bent portions 116 in two locations on the Y1 side of the first plate 11 are fitted to the recess portions 154 of the first coil holding portion 151, and the claw portions 155 are fitted to the notch portion 117 of each fixing bent portion 116 (see FIG. 9). As a result, the fixing bent portions 116 in two locations are locked to the side surfaces on both sides of the first coil holding portion 151 in the X direction.

Similarly, the fixing bent portions 116 in two locations on the Y2 side are fitted to the recess portions 164 of the second coil holding portion 161, and the claw portions 165 are fitted to the notch portion 117 of each fixing bent portion 116 (see FIG. 9). As a result, the fixing bent portions 116 in two locations are locked to the side surfaces on both sides of the second coil holding portion 161 in the X direction. As a result, the first holder member 15 and the second holder member 16 are positioned with respect to the first plate 11.

Next, the coil 10 is disposed between the first holder member 15 and the second holder member 16 assembled on the first plate 11, and the central hole 10c of the coil 10 is filled with an adhesive. Furthermore, the two coil wires 10d that have been drawn out from the coil 10 are drawn out from the notch portion 156 of the first holder side plate portion 152 to the Y1 side. Then, the second plate 12 is placed thereon from the Z2 side and the adhesive is cured. As shown in FIGS. 3 and 4, an adhesive layer 18 of the cured resin is formed in the central hole 10c of the coil 10. As a result, the second plate 12 is fixed to the coil 10 and the first plate 11 via the adhesive layer 18. The adhesive filling the central hole 10c enters the gap between the coil and the first coil holding portion 151 and the gap between the coil and the second coil holding portion 161 and hardens (not illustrated). This completes the coil assembly 13.

The width of the one side plate portion 123 and the other side plate portion 124 of the second plate 12 in the X direction is larger than that of the one side plate portion 113 and the other side plate portion 114 of the first plate 11. Therefore, when the second plate 12 is placed on the coil 10, the first coil holding portion 151, and the second coil holding portion 161 from the Z2 side, as shown in FIGS. 5, 6, and 9, the first coil holding portion 151 is fitted between the fixing bent portions 126 in two locations on the Y1 side of the second plate 12, and the second coil holding portion 161 is fitted between the fixing bent portions 126 in two locations on the Y2 side. Because the fixing bent portions 116 of the first plate 11 are fitted to the recess portions 154 of the first coil holding portion 151 and the recess portions 164 of the second coil holding portion 161, they are covered from the outside by the fixing bent portions 126 of the second plate 12.

When the actuator 1 is manufactured, the coil 10, the first plate 11, the second plate 12, the first holder member 15, and the second holder member 16 are assembled as described above to complete the coil assembly 13. Further, the yoke 8 and the magnet 7 are assembled so as to surround the coil assembly 13 to complete the movable body 5, and the movable body 5 and the coil assembly 13 are connected by the connecting body 4.

Then, the first case member 30 and the second case member 40 are assembled with respect to the coil assembly 13 to accommodate the movable body 5 in the case 2. The power feed substrate 14 may be attached to the coil assembly 13 before assembling the movable body 5, or may be attached after assembling the case 2.

Yoke

The yoke 8 is made of a magnetic material. As shown in FIGS. 3 and 4, the yoke 8 includes a first opposing portion 801 that opposes the coil 10 from the Z1 direction, and a second opposing portion 802 that opposes the coil 10 from the Z2 direction. The first magnet 71 is fixed to the first opposing portion 801. The second magnet 72 is fixed to the second opposing portion 802. Furthermore, as shown in FIG. 4, the yoke 8 is provided with a pair of connection portions 803 that extend in the Z direction from both sides in the X direction of the coil 10. The pair of connection portions 803 connects the first opposing portion 801 and the second opposing portion 802.

When the yoke 8 is assembled, a pair of second joining plate portions 805 that extends in the Z1 direction from both ends of the second opposing portion 802 in the X direction is press-fitted and fixed inside a pair of first joining plate portions 804 that extends in the Z2 direction from both ends of the first opposing portion 801 in the X direction. As a result, the pair of connection portions 803 is formed, and the yoke 8 is assembled in a shape that surrounds the outer peripheral side of the coil 10, the first plate 11, and the second plate 12 (see FIGS. 2 and 4).

As shown in FIGS. 5 and 6, the yoke 8 includes a first yoke 81 and a second yoke 82. The first yoke 81 is configured by two members that are joined together, namely a first inner member 83 that overlaps the coil 10 from the Z1 direction, and a first outer member 84 that overlaps the first inner member 83 from the Z1 direction. The second yoke 82 is configured by two members that are joined together, namely a second inner member 85 that overlaps the coil from the Z1 direction, and a second outer member 86 that overlaps the second inner member 85 from the Z1 direction.

The first opposing portion 801 of the yoke 8 is configured by stacking the first outer member 84 and the first inner member 83 in the Z direction. As shown in FIGS. 5 and 6, the pair of first joining plate portions 804 is provided on the first outer member 84, and extend toward the Z2 direction on both sides of the first inner member 83 and the first magnet 71 in the X direction. The first opposing portion 801 includes a pair of first connecting body fixing portions 806 that extend toward both sides of the first inner member 83 and the first magnet 71 in the Y direction. Each of the pair of first connecting body fixing portions 806 is connected to the first plate 11 via the first connecting bodies 9A.

The second opposing portion 802 of the yoke 8 is configured by stacking the second outer member 86 and the second inner member 85 in the Z direction. As shown in FIGS. 5 and 6, the pair of second joining plate portions 805 is provided on the second outer member 86, and extend toward the Z1 direction on both sides of the second inner member 85 and the second magnet 72 in the X direction. The second opposing portion 802 includes a pair of second

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connecting body fixing portions **807** that extend toward both sides of the second inner member and the second magnet **72** in the Y direction. Each of the pair of second connecting body fixing portions **807** is connected to the second plate **12** via the second connecting bodies **9B**.

Connecting Body

As shown in FIGS. **3**, **5**, and **6**, the first connecting bodies **9A** and the second connecting bodies **9B** each have a rectangular parallelepiped shape that is long in the X direction. As shown in FIG. **3**, the first connecting bodies **9A** are positioned on the Z1 side of the coil **10**. The second connecting bodies **9B** are positioned on the Z2 side of the coil **10**. The first connecting bodies **9A** are disposed in two locations on the Y1 side and the Y2 side of the first magnet **71**, and are formed of two members having the same shape. The second connecting bodies **9B** are disposed in two locations on the Y1 side and the Y2 side of the second magnet **72**, and are formed of two members having the same shape. The first connecting bodies **9A** and the second connecting bodies **9B** each exhibit at least one of elasticity and viscoelasticity.

As described above, the first connecting bodies **9A** are sandwiched between the first opposing portion **801** of the yoke **8** and the first plate **11** on both sides of the coil **10** in the Y direction. The first connecting bodies **9A** are compressed between the first opposing portion **801** and the first plate **11** in the Z direction. As described above, the second connecting bodies **9B** are sandwiched between the second opposing portion **802** of the yoke **8** and the second plate **12** on both sides of the coil **10** in the Y direction. The second connecting bodies **9B** are compressed between the second opposing portion **802** and the second plate **12** in the Z direction.

In the present embodiment, the first connecting bodies **9A** and the second connecting bodies **9B** are gel-like members made from silicone gel. Silicone gel is a viscoelastic body whose spring constant when deformation occurs in an expanding/contracting direction is approximately three times greater than the spring constant when deformation occurs in a shear direction. When a viscoelastic body is deformed in the direction that intersects a thickness direction (shear direction), because this is a deformation in a direction in which the viscoelastic body is pulled and stretched, the viscoelastic body has deformation characteristics in which the linear component is larger than the nonlinear component. Furthermore, when a viscoelastic body is subjected to compression deformation by being pressed in the thickness direction, it has expansion/contraction characteristics in which the nonlinear component is larger than the linear component, but on the other hand, when the viscoelastic body is pulled and stretched in the thickness direction, it has expansion/contraction characteristics in which the linear component is larger than the nonlinear component.

Alternatively, the first connecting bodies **9A** and the second connecting bodies **9B** may be formed using various rubber materials such as natural rubber, diene rubber (such as styrene-butadiene rubber, isoprene rubber, butadiene rubber, chloroprene rubber, and acrylonitrile-butadiene rubber), non-diene rubber (such as butyl rubber, ethylene-propylene rubber, ethylene-propylene-diene rubber, urethane rubber, silicone rubber, and fluoro-rubber), thermoplastic elastomers, and modified materials of these rubber materials.

FIG. **10** is a cross-sectional view of the actuator **1** showing the plan view shape and arrangement of the connecting body **4**, and is a cross-sectional view taken along position D-D in FIG. **3**. As shown in FIG. **10**, the second connecting bodies **9B** are connected to the one side plate

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portion **123** and the other side plate portion **124** of the second plate **12**. As shown in FIG. the length of the second connecting bodies **9B** is larger than the width of the central plate portion **122** in the X direction, and larger than the width of the coil **10**, which is covered by the central plate portion **122**, in the X direction.

The first connecting bodies **9A** have the same shape as the second connecting bodies **9B**, and are disposed in the same positions as the second connecting bodies **9B** when viewed from the Z direction. The first connecting bodies **9A** are connected to the one side plate portion **113** and the other side plate portion **114** of the first plate **11**. The length of the first connecting bodies **9A** is larger than the width of the central plate portion **112** in the X direction, and larger than the width of the coil **10**, which is covered by the central plate portion **112**, in the X direction.

Detailed Shape of Yoke

FIG. **11** is a side view of the movable body **5**, the connecting body **4**, and the coil assembly **13** when viewed from the X2 direction. As described above, in the yoke **8**, both end parts of the first opposing portion **801** and the second opposing portion **802** in the Y direction protrude on the Y1 side and the Y2 side of the connection portions **803**. Further, the first connecting body fixing portions **806** and the first connecting bodies **9A** are disposed on the Z1 side of the connection portions **803**, and the second connecting body fixing portions **807** and the second connecting bodies **9B** are disposed on the Z2 side of the connection portions **803**. That is, as shown in FIGS. **10** and **11**, in the yoke **8**, both sides of the pair of connection portions **803** in the Y direction are notched in a shape that exposes the first connecting bodies **9A** and the second connecting bodies **9B** when viewed from the X direction. As shown in FIG. **10**, the first connecting bodies **9A** and the second connecting bodies **9B** oppose the first holder projection portions **153** of the first holder member **15** and the second holder projection portions **163** of the second holder member **16** in the X direction.

When the movable body **5** is assembled around the coil assembly **13**, the connection portions **803** of the yoke **8** is disposed in the space between the leading edges of the first holder projection portions **153** in the Y2 direction and the leading edges of the second holder projection portions **163** in the Y1 direction (see FIGS. **2** and **9**). Here, each of the pair of connection portions **803** is provided with a first notch portion **808**, in which a central part in the Z direction of the edge on the Y1 side is notched toward the Z2 side, and a second notch portion **809**, in which a central part in the Z direction of the edge on the Y2 side is notched toward the Y1 side (see FIGS. **2** and **11**). As shown in FIG. **11**, when viewed from the X direction, the end portion of the first coil holding portion **151** on the Y2 side is disposed in the first notch portion **808**, and the end portion of the second coil holding portion **161** on the Y1 side is disposed in the second notch portion **809**. That is, in the yoke **8**, both end portions of the pair of connection portions **803** in the Y direction are notched in a shape that does not interfere with first coil holding portion **151** and the second coil holding portion **161**.

Furthermore, when the movable body **5** is assembled around the coil assembly **13**, as shown in FIG. **2**, the first connecting body fixing portion **806** provided on the end portion of the first opposing portion **801** on the Y1 side and the second connecting body fixing portion **807** provided on the end portion of the second opposing portion **802** on the Y1 side are disposed between the pair of first holder projection portions **153** that are opposed in the X direction. In addition, the first connecting body fixing portion **806** provided on the end portion of the first opposing portion **801** on

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the Y2 side and the second connecting body fixing portion **807** provided on the end portion of the second opposing portion **802** on the Y2 side are disposed between the pair of second holder projection portions **163** that are opposed in the X direction. Therefore, the first holder projection portions **153** and the second holder projection portions **163** of the coil holder **17** collide with the first connecting body fixing portions **806** and the second connecting body fixing portions **807** and act as stoppers that regulate excessive movement of the movable body **5** when the movable body **5** moves significantly in the X direction due to an impact or the like.

Case

As shown in FIGS. 1A, 1B, and 2, the first case member **30** includes a first end plate portion **31** having a substantially rectangular shape that opposes the coil **10** from the Z1 direction. Furthermore, the first case member **30** includes a first case first side plate portion **32** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y1 direction, a first case second side plate portion **33** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y2 direction, a first case third side plate portion **34** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X1 direction, and a first case fourth side plate portion **35** that extends toward the Z2 direction from the end of the first end plate portion **31** in the X2 direction.

The edges of the first end plate portion **31** in the X direction are provided with projection parts in a central part in the Y direction that protrude on both sides in the X direction. As a result, the first case third side plate portion **34** and the first case fourth side plate portion **35** are each provided with a side plate central portion **301** that extends from the leading edge of the projection parts on the first end plate portion **31** toward the Z1 direction, and side plate end parts **302** that are provided in positions shifted further toward the inside (central side of the first case member **30** in the X direction) than the side plate central portion **301**, in both sides of the side plate central portion **301** in the Y direction. Both ends of the side plate central portion **301** in the Y direction are connected to a step portion that is substantially bent in a right angle toward the inside. The side plate central portion **301** and the side plate end parts **302** are connected via the step portions.

As shown in FIG. 2, the first case first side plate portion **32** has a shape that does not interfere with the substrate fixing portion **159** provided on the first holder side plate portion **152**, and is notched in a shape that surrounds the Z1 side, the X1 side, and the X2 side of the substrate fixing portion **159**. The first case second side plate portion **33** is provided with circular first projection portions **36** that protrude toward the inside (Y1 direction) of the first case member **30**. The first projection portions **36** are half punched portions formed by deforming a metal plate into a shape that protrudes in the Y1 direction by press working. Therefore, the first projection portions **36** have a surface on the Y2 side with a concave shape that is recessed toward the Y1 side, and have a surface on the Y1 side with a convex shape that protrudes toward the Y1 side. The first projection portions **36** are formed in a leading edge part of the first case second side plate portion **33** in the Z2 direction in two locations that are separated in the X direction.

In the present embodiment, when the first case member **30** is assembled with respect to the coil assembly **13**, the coil holder **17** is fitted between the first case second side plate portion **33** and the first case first side plate portion **32**. At this time, as shown in the partially enlarged view shown in FIG.

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9, the first projection portions **36** provided in two locations on the first case second side plate portion **33** make contact with the second holder side plate portion **162**, which is disposed on the end portion of the coil holder **17** on the Y2 side. The first projection portions **36** make elastic contact with the first case second side plate portion **33** from the Y2 side and press the coil holder **17** to the Y1 side. As a result, the coil holder **17** is biased in the Y1 direction inside the first case member **30**, and the first holder side plate portion **152** provided on the end portion of the coil holder **17** on the Y1 side is pressed against the first case first side plate portion **32** from the Y2 side (see FIGS. 3 and 9). Consequently, the coil holder **17** is positioned with respect to the first case member **30** in a state where it does not rattle in the Y direction.

The substrate fixing portion **159** provided on the coil holder **17** protrudes on the Y1 side from the notch provided in the first case first side plate portion **32** as described above. As shown in FIG. 3, in the present embodiment, the side surface of the first holder side plate portion **152** in the Y1 direction (contact surface **158**) is making contact with the first case first side plate portion **32**, and the step between the substrate fixing portion **159** and the contact surface **158** is larger than the plate thickness of the first case first side plate portion **32**. As a result, the end portion of the power feed substrate **14** on the Z1 side, which extends toward the Z1 side of the substrate fixing portion **159**, does not make contact with the first case first side plate portion **32**, and a space S (see FIG. 3) in the Y direction is formed between the power feed substrate **14** and the first case first side plate portion **32**. Therefore, it is possible to prevent the pattern on the power feed substrate **14** from being short-circuited due to contact with the case **2**.

Next, the second case member **40** includes a second end plate portion **41** having a substantially rectangular shape that opposes the coil **10** from the Z2 direction. Furthermore, the second case member **40** includes a second case first side plate portion **42** that extends toward the Z1 direction from the end of the second end plate portion **41** in the Y1 direction, a second case second side plate portion **43** that extends toward the Z1 direction from the end of the second end plate portion **41** in the Y2 direction, a second case third side plate portion **44** that extends toward the Z1 direction from the end of the second end plate portion **41** in the X1 direction, and a second case fourth side plate portion **45** that extends toward the Z1 direction from the end of the second end plate portion **41** in the X2 direction.

The edges of the second end plate portion **41** in the X direction are provided with projection parts in a central part in the Y direction that protrudes on both sides in the X direction. As a result, the second case third side plate portion **44** and the second case fourth side plate portion **45** are each provided with a side plate central portion **401** that extends from the leading edge of the projection parts on the second end plate portion **41** toward the Z2 direction, and side plate end parts **402** that are provided on both sides of the side plate central portion **401** in positions shifted further toward the inside (central side of the second case member **40** in the X direction) than the side plate central portion **401**, in both sides of the side plate central portion **401** in the Y direction. Both ends of the side plate central portion **401** in the Y direction are connected to a step portion that is substantially bent in a right angle toward the inside. The side plate central portion **401** and the side plate end parts **402** are connected via the step portions.

The first case member **30** and the second case member **40** are positioned as a result of the pair of first side plate portions provided on both ends of the first case member **30**

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in the X direction (that is, the first case third side plate portion **34** and the first case fourth side plate portion **35**) being press-fitted between the pair of second side plate portions provided on both ends of the second case member **40** in the X direction (that is, the second case third side plate portion **44** and the second case fourth side plate portion **45**). After being temporarily fixed by press-fitting, the first case member **30** and the second case member **40** are fully fixed by welding the location where the second case third side plate portion **44** and the first case third side plate portion **34** overlap each other and the location where the second case fourth side plate portion **45** and the first case fourth side plate portion **35** overlap each other.

As shown in FIG. 2, the second case third side plate portion **44** is provided with second projection portions **46** that protrude in the X2 direction (that is, toward the inside of the second case member **40**). Similarly, the second case fourth side plate portion **45** is provided with third projection portions **47** that protrude in the X1 direction (that is, toward the inside of the second case member **40**). The second projection portions **46** and the third projection portions **47** are formed in two locations on both ends of the side plate central portion **401** in the Y direction, and formed in one location on each of the side plate end parts **402** provided on both sides of the side plate central portion **401**. The second projection portions **46** and the third projection portions **47** are half punched portions having the same shape as the first projection portions **36**.

The second projection portions **46** and the third projection portions **47** are formed in opposing positions in the X direction. The pair of first side plate portions (the first case third side plate portion **34** and the first case fourth side plate portion **35**) of the first case member **30** are press-fitted between the opposing second projection portions **46** and third projection portions **47**. As shown in the partially enlarged view of FIG. 9, the second projection portions **46** make elastic contact with the first case third side plate portion **34** from the X1 side. Furthermore, the third projection portions **47** make elastic contact with the first case fourth side plate portion **35** from the X2 side.

As described below, in the present embodiment, because the coil holder **17** is press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**, a pressing force in directions expanding toward both sides in the X direction always acts on the first case third side plate portion **34** and the first case fourth side plate portion **35**. Therefore, when the first case member **30** and the second case member **40** are assembled, a state is formed in which a pressing force that presses outward from the first case third side plate portion **34** and the first case fourth side plate portion **35** always acts on the second case third side plate portion **44** and the second case fourth side plate portion **45**.
Press-Fitted Fixed Portion

The coil holder **17** is fixed to the case **2** as a result of the first holder member **15** and the second holder member **16** each being press-fitted inside the first case member **30**. As shown in FIGS. 2 and 9, the first holder member **15** includes a press-fitted fixed portion **50** that is press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**. Furthermore, the second holder member **16** includes a press-fitted fixed portion **60** that is press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**.

FIG. 12 is a cross-sectional view of the actuator **1** taken at the position of the press-fitted fixed portion **50** (a cross-sectional view taken along position E-E in FIG. 9). In the first holder member **15**, the first holder side plate portion **152**

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and the first holder projection portions **153** constitute the press-fitted fixed portion **50**. As shown in FIGS. 9 and 12, the press-fitted fixed portion **50** is press-fitted between the pair of side plate end parts **302** provided on the end portion of the first case third side plate portion **34** and the first case fourth side plate portion **35** in the Y1 direction.

As shown in FIGS. 9 and 12, the end portion of the press-fitted fixed portion **50** in the X1 direction (that is, the first holder projection portion **153** on the X1 side) includes a one side fixing surface **51** that makes contact with the side plate end part **302** of the first case third side plate portion **34** from the inside. The end portion of the press-fitted fixed portion **50** in the X2 direction (that is, the first holder projection portion **153** on the X2 side) includes an other side fixing surface **52** that makes contact with the side plate end part **302** of the first case fourth side plate portion **35** from the inside.

As shown in FIGS. 2, 7 and 12, the one side fixing surface **51** and the other side fixing surface **52** is provided on a central part of the press-fitted fixed portion **50** in the Z direction. The Z1 side and the Z2 side of the one side fixing surface **51** and the other side fixing surface **52** are each provided with step portions in the X direction. That is, the end portion of the press-fitted fixed portion **50** in the X1 direction is provided with a one side recess portion **53** on the Z1 side of the one side fixing surface **51** that is more recessed than the one side fixing surface **51** in the X2 direction, and a one side projection portion **54** on the Z2 side of the one side fixing surface **51** that protrudes further in the X1 direction than the one side fixing surface **51**. Similarly, the end portion of the press-fitted fixed portion **50** in the X2 direction is provided with an other side recess portion **55** on the Z1 side of the other side fixing surface **52** that is more recessed than the other side fixing surface **52** in the X1 direction, and an other side projection portion **56** on the Z2 side of the other side fixing surface **52** that protrudes further in the X2 direction than the other side fixing surface **52**.

As shown in FIG. 12, in the press-fitted fixed portion **50**, the one side fixing surface **51** and the other side fixing surface **52** provided in a central part in the Z direction make contact with the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35** in the Z2 direction, and the parts in the Z1 direction (the one side recess portion **53** and the other side recess portion **55**) do not make contact with the first case third side plate portion **34** and the first case fourth side plate portion **35**. Therefore, the press-fitted fixed portion **50** presses the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35**, which are in the Z2 direction, in directions expanding toward the outside (both sides in the X direction).

The press-fitted fixed portion **50** is positioned in the Z direction with respect to the first case member **30** as a result being press-fitted to a position in which the leading edges of the first case third side plate portion **34** and the first case fourth side plate portion **35** each make contact with a step portion between the one side projection portion **54** and the one side fixing surface **51** and a step portion between the other side projection portion **56** and the other side fixing surface **52**. The end portions of the press-fitted fixed portion **50** in the Z2 direction (the one side projection portion **54** and the other side projection portion **56**) are covered from the outside (both sides in the X direction) by the second case third side plate portion **44** and the second case fourth side plate portion **45** of the second case member **40**.

Next, in the second holder member **16**, the second holder side plate portion **162** and the second holder projection

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portions **163** constitute the press-fitted fixed portion **60** (see FIGS. **2**, **7** and **9**). The press-fitted fixed portion **60** is press-fitted between the pair of side plate end parts **302** provided on the end portion of the first case third side plate portion **34** and the first case fourth side plate portion **35** in the Y2 direction.

The press-fitted fixed portion **60** is configured in the same manner as the press-fitted fixed portion **50**. That is, the end portion of the press-fitted fixed portion **60** in the X1 direction includes a one side fixing surface **61** that makes contact with the side plate end part **302** of the first case third side plate portion **34** from the inside, a one side recess portion **63** on the Z1 side of the one side fixing surface **61** that is more recessed than the one side fixing surface **61** in the X2 direction (see FIG. **6**), and a one side projection portion **64** on the Z2 side of the one side fixing surface **61** that protrudes further in the X1 direction than the one side fixing surface **61**. Furthermore, the end portion of the press-fitted fixed portion **60** in the X2 direction is provided with an other side fixing surface **62** that makes contact with the side plate end part **302** of the first case fourth side plate portion **35** from the inside, an other side recess portion **65** on the Z1 side of the other side fixing surface **62** that is more recessed than the other side fixing surface **62** in the X1 direction, and an other side projection portion **66** on the Z2 side of the other side fixing surface **62** that protrudes further in the X2 direction than the other side fixing surface **62**.

In the press-fitted fixed portion **60**, the one side fixing surface **61** and the other side fixing surface **62** provided in a central part in the Z direction make contact with the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35**, and the end portions in the Z1 direction (the one side recess portion **63** and the other side recess portion **65**) do not make contact with the first case third side plate portion **34** and the first case fourth side plate portion **35**. Therefore, the press-fitted fixed portion **60** presses the leading edge parts of the first case third side plate portion **34** and the first case fourth side plate portion **35**, which are in the Z2 direction, in directions expanding toward the outside (both sides in the X direction).

Furthermore, the press-fitted fixed portion **60** is positioned in the Z direction with respect to the first case member **30** as a result being press-fitted to a position in which the leading edges of the first case third side plate portion **34** and the first case fourth side plate portion **35** each make contact with a step portion between the one side projection portion **64** and the one side fixing surface **61** and a step portion between the other side projection portion **66** and the other side fixing surface **62**. The end portions of the press-fitted fixed portion **60** in the Z2 direction (the one side projection portion **64** and the other side projection portion **66**) are covered from the outside (both sides in the X direction) by the second case third side plate portion **44** and the second case fourth side plate portion **45** of the second case member **40**.

As a result of such a fixing structure, the pair of first side plate portions (the first case third side plate portion **34** and the first case fourth side plate portion **35**) of the first case member are always pressed toward the outside by a pressing force from the press-fitted fixed portions **50** and **60**. Therefore, a state is formed in which the first case third side plate portion **34** and the first case fourth side plate portion **35** are always pressed from the inside against the pair of second side plate portions (the second case third side plate portion **44** and the second case fourth side plate portion **45**) of the second case member **40** that has been placed on the first case

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member **30**. Therefore, the first case member **30** is positioned with respect to the second case member **40** in a state where it does not rattle.

Main Effects of Present Embodiment

As described above, the actuator **1** of the present embodiment includes: the movable body **5**; the support body **3** provided with the metallic case **2** that accommodates the movable body **5**, and the coil holder **17** made of resin; the connecting body **4** connected to the movable body **5** and the support body **3**; and the magnetic drive circuit **6** which includes the coil **10** held by coil holder **17** and the magnet **7** that opposes the coil **10** in the Z direction (first direction), and causes the movable body **5** to vibrate with respect to the support body **3** in the X direction (second direction). The case includes the first case member **30**, which is provided with the first end plate portion **31** that opposes the movable body **5** from the Z1 direction, and the second case member which is provided with the second end plate portion **41** that opposes the movable body **5** from the Z2 direction. The first case member **30** includes the first case first side plate portion **32** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y1 direction, and the first case second side plate portion **33** that extends toward the Z2 direction from the end of the first end plate portion **31** in the Y2 direction. The first case second side plate portion **33** is provided with the first projection portion **36** that protrudes in the Y1 direction, and the coil holder **17** is press-fitted between the first projection portion **36** and the first case first side plate portion **32**.

In the present embodiment, one of the side plates of the first case member **30** that are opposed in the Y direction (first case second side plate portion **33**) is provided with a projection shape (first projection portion **36**). Therefore, when the case **2** and the coil holder **17** are assembled, the first projection portion **36** can be brought into elastic contact with the coil holder **17**. Further, the coil holder **17** can be positioned as a result of being biased toward the other side plate (first case first side plate portion **32**). As a result, rattling of the coil holder **17** inside the case can be suppressed, and the noise due to rattling of the coil holder **17** can be suppressed. Moreover, even when the dimensional accuracy of the first case member **30** is low, the coil holder **17** can be brought into contact with the first case first side plate portion **32** with certainty, which provides good ease of assembly. In addition, because it is not necessary to fix the coil holder **17** to the case **2** using an adhesive, an injection step and curing step of the adhesive is not necessary. Therefore, an increase in the number of assembly steps can be suppressed.

In the present embodiment, in the first projection portion **36**, the surface in the Y2 direction has a concave shape, and the surface in the Y1 direction is a half punched portion having a convex shape. Therefore, the first case member **30** provided with the first projection portion **36** can be easily manufactured because it can be easily formed by a method such as press working. Therefore, an increase in component costs can be suppressed.

In the present embodiment, the coil holder **17** is provided with: the first holder member which includes the first coil holding portion **151** disposed in the Y1 direction with respect to the coil **10**, and the first holder side plate portion **152** that extends in the Z1 direction and the Z2 direction from the end of the first coil holding portion **151** in the Y1 direction; and the second holder member **16**, which includes the second coil holding portion **161** disposed in the Y2

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direction with respect to the coil 10, and the second holder side plate portion 162 that extends in the Z1 direction and the Z2 direction from the end of the second coil holding portion 161 in the Y2 direction. The first holder member 15 and the second holder member 16 are fixed via the metallic first plate 11 that overlaps the first coil holding portion 151 and the second coil holding portion 161 from the Z1 direction, and the metallic second plate 12 that overlaps the first coil holding portion 151 and the second coil holding portion 161 from the Z2 direction. The first projection portion 36 makes contact with the second holder side plate portion 162 from the Y2 direction. The first holder side plate portion 152 is biased toward the Y1 direction inside the first case member 30 and makes contact with the first case first side plate portion 32 from the Y2 direction. As a result, by dividing the coil holder 17 into two members by dividing it in the Y direction, a resin part that covers both sides of the coil 10 in the X direction becomes unnecessary. Therefore, the external dimensions of the actuator 1 in the X direction can be reduced.

Alternatively, the coil 10 can be made larger without an increase in the outer shape of the actuator 1, and the thrust of the magnetic drive circuit 6 can be increased to generate large vibrations. Furthermore, because the coil 10 can be protected by the first plate 11 and the second plate 12, the possibility that the coil 10 will be damaged due to collision with the magnet 7 or collision with the yoke 8 is low. In addition, the first holder member 15 and the second holder member 16 can be handled as the coil assembly 13 as a result of being fixed via the first plate 11 and the second plate 12. Therefore, even when the coil holder 17 is divided into two members, a reduction in the ease of assembly of the actuator 1 can be suppressed, and it is also possible for the first projection portion 36 to be brought into elastic contact and positioned with respect to the coil holder 17.

In the present embodiment, the coil holder 17 includes the contact surface 158 that makes contact with the first case first side plate portion 32 from the Y2 direction, and the substrate fixing portion 159 that protrudes from the contact surface 158 toward the Y1 direction of the first case first side plate portion 32. As a result, a space S is formed in the Y direction (third direction) between the power feed substrate 14 that is fixed to the substrate fixing portion 159, and the first case first side plate portion 32. Therefore, because contact between the metallic case 2 and the power feed substrate 14 can be avoided, it is possible to prevent the pattern on the power feed substrate 14 and the case 2 from making contact and short-circuiting.

In the present embodiment, the first case member 30 includes the first case third side plate portion 34 that extends toward the Z2 direction from the end of the first end plate portion 31 in the X1 direction, and the first case fourth side plate portion 35 that extends toward the Z2 direction from the end of the first end plate portion 31 in the X2 direction. The second case member 40 includes the second case third side plate portion 44 that extends toward the Z1 direction from the end of the second end plate portion 41 in the X1 direction, and the second case fourth side plate portion 45 that extends toward the Z1 direction from the end of the second end plate portion 41 in the X2 direction. The first case third side plate portion 34 and the first case fourth side plate portion 35 are press-fitted between the second case third side plate portion 44 and the second case fourth side plate portion 45. In such a configuration where the first case member 30 and the second case member 40 can be posi-

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tioned by press-fitting, they do not need to be fixed using an adhesive. Therefore, an increase in the number of assembly steps can be suppressed.

In the present embodiment, the second case third side plate portion 44 is provided with the second projection portion 46 that protrudes toward the first case third side plate portion 34, and the second case fourth side plate portion 45 is provided with the third projection portion 47 that protrudes toward the first case fourth side plate portion 35. As a result, a press-fitted state can be formed by bringing the leading edge of the second projection portion 46 into elastic contact with the first case third side plate portion 34, and bringing the leading edge of the third projection portion 47 into elastic contact with the first case fourth side plate portion 35. Therefore, press-fitting can be performed with certainty even when the dimensional accuracy of the first case member 30 and the second case member 40 is low, and a reduction in the ease of assembly can also be suppressed.

In the present embodiment, the second projection portion 46 and the third projection portion 47 are half punched portions. That is, in the second projection portion 46, the surface in the X1 direction has a concave shape, and the surface in the X2 direction is a half punched portion having a convex shape. Furthermore, in the third projection portion 47, the surface in the X2 direction has a concave shape, and the surface in the X1 direction is a half punched portion having a convex shape. Therefore, the second case member 40 provided with the second projection portion 46 and the third projection portion 47 can be easily manufactured because it can be easily formed by a method such as press working. Therefore, an increase in component costs can be suppressed.

In the present embodiment, the movable body 5 is provided with the yoke 8 which holds the magnet 7. The yoke 8 includes the first opposing portion 801 that opposes the coil 10 from the Z1 direction, the second opposing portion 802 that opposes the coil 10 from the Z2 direction, and the pair of connection portions 803 that extend in the Z direction on both sides of the coil 10 in the X direction and connect the first opposing portion 801 and the second opposing portion 802. The coil holder 17 includes the pair of first holder projection portions 153 that protrude toward the Y2 direction from the edges on both sides of the first holder side plate portion 152 in the X direction, and the pair of second holder projection portions 163 that protrude toward the Y1 direction from the edges on both sides of the second holder side plate portion 162 in the X direction. The end portions of the first opposing portion 801 and the second opposing portion 802 in the Y1 direction are disposed between the pair of first holder projection portions 153. The end portions of the first opposing portion 801 and the second opposing portion 802 in the Y2 direction are disposed between the pair of second holder projection portions 163. As a result, the first holder projection portions 153 and the second holder projection portions 163 function as stoppers that regulate the movable range of the movable body 5 in the X direction. In this way, by providing stoppers on the coil holder 17 instead of the case 2, it is less likely that the movable body 5 will collide with the case and be deformed or destroyed due to an impact caused by falling or the like. Therefore, the impact resistance of the actuator 1 can be increased.

In the present embodiment, the first case member 30 includes the first case third side plate portion 34 that extends toward the Z2 direction from the end of the first end plate portion 31 in the X1 direction, and the first case fourth side plate portion 35 that extends toward the Z2 direction from the end of the first end plate portion 31 in the X2 direction.

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The pair of first holder projection portions **153** and the pair of second holder projection portions **163** are press-fitted between the first case third side plate portion **34** and the first case fourth side plate portion **35**. In this way, by press-fitting the coil holder **17** between the side plates (the first case third side plate portion **34** and the first case fourth side plate portion **35**) that are opposed in the X direction, rattling of the coil holder **17** in the X direction can be suppressed. Furthermore, the first case third side plate portion **34** and the first case fourth side plate portion **35** are in a state where they are always pressed toward the outside by the coil holder **17**. As a result, because the first case third side plate portion **34** and the first case fourth side plate portion **35** are always pressed against the second case third side plate portion **44** and the second case fourth side plate portion **45** from the inside, the first case member **30** can be positioned with respect to the second case member **40** in a state where it does not rattle. Moreover, because the parts that function as stoppers that regulate the movable range of the movable body **5** in the X direction can be utilized for press-fitted fixing, the press-fitted fixing can be firmly performed and the configuration of the coil holder **17** can be simplified.

Modifications

(1) In the above embodiment, the first connecting bodies **9A** and the second connecting bodies **9B** are provided as the connecting body **4**. However, a configuration is possible in which only one of the first connecting bodies **9A** and the second connecting bodies **9B** is provided.

(2) In the above embodiment, the first magnet **71** and the second magnet **72** are provided as the magnet **7**. However, a configuration is possible in which only one of the first magnet **71** and the second magnet **72** are provided.

(3) In the above embodiment, in the yoke **8**, the first yoke **81** and the second yoke **82** are each configured stacking an inner member and an outer member. However, the first yoke **81** and the second yoke **82** may each be configured by only an outer member.

(4) In the above embodiment, the yoke **8** has the pair of second joining plate portions **805** press-fitted inside the pair of first joining plate portion **804**. However, a configuration is also possible in which the pair of first joining plate portions **804** is press-fitted inside the pair of second joining plate portions **805**.

(5) In the above embodiment, the coil holder **17** is split into two parts in the Y direction and is configured by two members. However, the coil holder **17** may be configured by a single member. For example, the coil holder **17** can be configured having a coil placement hole provided in a plate portion connecting the first holder side plate portion **152** and the second holder side plate portion **162**.

(6) The above embodiment has a configuration in which projection portions (the second projection portions **46** and the third projection portions **47**) that protrude toward the inside are provided on the side plates of the second case member **40** in the X direction (the second case third side plate portion **44** and the second case fourth side plate portion **45**). However, the projection portions may be provided on either of the first case member **30** and the second case member **40**. That is, the second projection portions that protrude toward the outside may be provided on the first case third side plate portion **34**, and the third projection portions that protrude toward the outside may be provided on the first case fourth side plate portion **35**.

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What is claimed is:

1. An actuator comprising:

a movable body;

a support body provided with a metallic case that accommodates the movable body, and a coil holder made of resin;

a connecting body that is connected to the movable body and the support body; and

a magnetic drive circuit which is provided with a coil that is held by the coil holder, and a magnet that opposes the coil in a first direction, and which vibrates the movable body with respect to the support body in a second direction that intersects the first direction; wherein

the case is provided with a first case member having a first end plate portion that opposes the movable body from one side in the first direction, and a second case member having a second end plate portion that opposes the movable body from an other side in the first direction,

the first case member is provided with a first case first side plate portion that extends toward the other side in the first direction from an end on one side of the first end plate portion in a third direction, and a first case second side plate portion that extends toward the other side in the first direction from an end on an other side of the first end plate portion in the third direction, the third direction being a direction that intersects the first direction and intersects the second direction,

the first case second side plate portion is provided with a first projection portion that protrudes toward the one side in the third direction, and

the coil holder is press-fitted between the first projection portion and the first case first side plate portion.

2. The actuator according to claim 1, wherein in the first projection portion, a surface on the other side in the third direction has a concave shape, and a surface on the one side in the third direction is a half punched portion having a convex shape.

3. The actuator according to claim 1, wherein the coil holder is provided with a contact surface that makes contact with the first case first side plate portion from the other side in the third direction, and a substrate fixing portion that protrudes from the contact surface toward the one side of the first case first side plate portion in the third direction.

4. The actuator according to claim 1, wherein the coil holder comprises

a first holder member provided with a first coil holding portion disposed on the one side of the coil in the third direction, and a first holder side plate portion that extends in the first direction from an end of the first coil holding portion on the one side in the third direction, and

a second holder member provided with a second coil holding portion disposed on the other side of the coil in the third direction, and a second holder side plate portion that extends in the first direction from an end of the second coil holding portion on the other side in the third direction,

the first holder member and the second holder member are fixed via a metallic first plate, which overlaps the first coil holding portion and the second coil holding portion from the one side in the first direction, and a metallic second plate, which overlaps the first coil holding portion and the second coil holding portion from the other side in the first direction,

the first projection portion makes contact with the second holder side plate portion from the other side in the third direction, and

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the first holder side plate portion makes contact with the first case first side plate portion from the other side in the third direction.

5. The actuator according to claim 4, wherein the movable body is provided with a yoke that holds the magnet,

the yoke is provided with a first opposing portion that opposes the coil from the one side in the first direction, a second opposing portion that opposes the coil from the other side in the first direction, and a pair of connection portions that extend in the first direction on both sides of the coil in the second direction and connect the first opposing portion and the second opposing portion,

the coil holder is provided with a pair of first holder projection portions that protrude toward the other side in the third direction from edges on both sides of the first holder side plate portion in the second direction, and a pair of second holder projection portions that protrude toward the one side in the third direction from edges on both sides of the second holder side plate portion in the second direction,

an end portion of the first opposing portion and the second opposing portion on the one side in the third direction is disposed between the pair of first holder projection portions, and

an end portion of the first opposing portion and the second opposing portion on the other side in the third direction is disposed between the pair of second holder projection portions.

6. The actuator according to claim 5, wherein

the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction, and

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the pair of first holder projection portions and the pair of second holder projection portions are press-fitted between the first case third side plate portion and the first case fourth side plate portion.

7. The actuator according to claim 1, wherein the first case member is provided with a first case third side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on one side in the second direction, and a first case fourth side plate portion that extends toward the other side in the first direction from an end of the first end plate portion on an other side in the second direction,

the second case member is provided with a second case third side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the one side in the second direction, and a second case fourth side plate portion that extends toward the one side in the first direction from an end of the second end plate portion on the other side in the second direction, and

the first case third side plate portion and the first case fourth side plate portion are press-fitted between the second case third side plate portion and the second case fourth side plate portion.

8. The actuator according to claim 7, wherein one of the second case third side plate portion and the first case third side plate portion is provided with a second projection portion that protrudes toward the other of the second case third side plate portion and the first case third side plate portion, and

one of the second case fourth side plate portion and the first case fourth side plate portion is provided with a third projection portion that protrudes toward the other of the second case fourth side plate portion and the first case fourth side plate portion.

9. The actuator according to claim 8, wherein the second projection portion and the third projection portion are half punched portions.

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